Stationary Reweighting Yields Local Convergence of Soft Fitted Q-Iteration

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Abstract

Fitted Q-iteration (FQI) and its entropy-regularized variant, soft FQI, are central tools for value-based offline reinforcement learning, but they can be unstable under function approximation and distribution shift. We show that, in the entropy-regularized setting, the soft Bellman operator is locally contractive in the stationary norm of the soft-optimal policy, rather than the behavior norm used by standard FQI. This mismatch helps explain why soft Q-iteration can fail under function approximation when Bellman completeness does not hold. To address this, we introduce *stationary-reweighted soft FQI*, which reweights each regression update by the stationary distribution of the current policy. Under approximate realizability, we prove local linear convergence, with weight-estimation errors geometrically damped across iterations. Our analysis yields a new treatment of approximate dynamic programming under local contraction. It also suggests that global convergence may be recovered via *temperature annealing*, by gradually lowering the softmax temperature, and that the approach extends to the hardmax limit under a mild margin condition.

1. Introduction

A core objective in reinforcement learning is to compute the optimal value function, defined as the fixed point of a Bellman optimality equation (Bellman, 1966). The hard $\max$ in this operator introduces nondifferentiability and can exacerbate instability when combined with function approximation and bootstrapping. Entropy regularization replaces the hard maximum with a smooth softmax, yielding a more stable "soft" Bellman operator that underlies many modern value-based algorithms (Ziebart et al., 2008; Ziebart, 2010; Haarnoja et al., 2017; 2018).

Fitted Q-iteration and its entropy-regularized analogue, *soft* FQI, approximate this fixed point by iterating a projected Bellman update: each iteration forms bootstrapped one-step Bellman targets and regresses them onto a function class. These algorithms are central in offline RL and large-scale approximate dynamic programming (Ernst et al., 2005; Munos, 2005; Lazaric et al., 2012; Uehara et al., 2023). When the function class is Bellman complete, classical analyses establish finite-sample guarantees and contraction-like behavior for the projected Bellman operator (Gordon, 1995; Tsitsiklis & Van Roy, 1996; Munos & Szepesvári, 2008; Scherrer, 2014; Fan et al., 2020).

However, fitted value-iteration methods such as FQI can degrade under function approximation and distribution shift when Bellman completeness fails (Chen & Jiang, 2019; Patterson et al., 2022). Recent information-theoretic and hardness results show that, without additional structural assumptions, offline value-function approximation can exhibit poor horizon dependence, exponential error amplification, or even loss of convergence (Chen & Jiang, 2019; Foster et al., 2021; Wang et al., 2021; Amortila et al., 2020; Meyn, 2024). Classical examples involving projected and bootstrapped operators (Gordon, 1995; Baird et al., 1995; Tsitsiklis & Van Roy, 1996) likewise demonstrate instability under function approximation. These limitations have motivated minimax and adversarial Q-learning methods that relax Bellman completeness through saddle-point formulations (Dai et al., 2018; Uehara et al., 2020; 2021; 2023; Jin et al., 2021; Xie & Jiang, 2021), but at the cost of more complex objectives, auxiliary critic classes, and a departure from the simplicity of standard supervised learning.

Motivated by these challenges, we identify a more basic source of instability: a *norm mismatch* between the regression update and the soft Bellman operator. Standard FQI performs regression under the behavior distribution, whereas the soft Bellman operator is contractive—*locally around its fixed point*—under the stationary distribution of the soft-optimal policy. This mismatch breaks the fixed-point geometry of dynamic programming and helps explain why standard FQI can perform poorly when Bellman completeness fails. It also suggests a simple remedy: reweight the regression step so that it is carried out in the stationary norm under which the Bellman operator contracts.

1.1. Contributions

Building on this geometric perspective, our contributions are:

1. **Stationary-weighted soft Q-iteration.** We define a projected soft Bellman operator in the stationary norm of the soft-optimal policy and show that it is locally contractive around the soft-optimal fixed point.

2. **A practical stationary-reweighted soft FQI algorithm.** We propose a fitted Q-iteration method that reweights each regression step using an estimate of the current policy's stationary density ratio.

3. **Finite-sample convergence guarantees.** We show that, within the contraction region, the iterates converge linearly, with density-ratio estimation error entering through a geometrically damped term. The key technical ingredient is a new analysis of approximate dynamic programming under locally contractive inexact Picard iteration.

4. **Weak structural assumptions.** Our analysis requires only approximate realizability, convexity of the function class, and the existence of an admissible weighting function, such as a stationary density ratio. In particular, it does not require Bellman completeness.

Scope and limitations. Our focus is primarily theoretical. In particular, the guarantees in this paper are inherently *local*: stationary-reweighted soft FQI is contractive only within a neighborhood of the target fixed point. We do not characterize when a practical algorithm enters this region from an arbitrary initialization, nor do we establish global convergence. Extending the theory in either direction remains an important open problem. One possible route is *temperature annealing*, which gradually lowers the softmax temperature; see Section 4 and Appendix B.

A second limitation is density-ratio estimation. Although there is a large literature on stationary density-ratio estimation (see Appendix A) and on emphatic weighting more broadly (Patterson et al., 2022), obtaining stable and accurate weights can be challenging in practice. Our goal here is not to solve that problem, but to isolate the role of stationary reweighting as a stabilizing mechanism for fitted soft Q-iteration.

1.2. Related Work

Most existing analyses of fitted value iteration assume realizability and (approximate) Bellman completeness. Under these assumptions, the projected Bellman operator is a global contraction in the sup norm (Munos, 2003; 2005; Munos & Szepesvári, 2008; Antos et al., 2007). Under function approximation, however, these assumptions often fail: projecting back onto a restricted function class can break the Bellman update and destroy this contraction property (Gordon, 1995; Baird et al., 1995; Tsitsiklis & Van Roy, 1996; Meyn, 2024; Lim & Lee, 2025; Patterson et al., 2022). As a result, errors can accumulate and, in offline settings, may even grow exponentially (Chen & Jiang, 2019; Foster et al., 2021; Wang et al., 2021; Amortila et al., 2020). We show that reweighting the Bellman regression step using stationary ratios restores a *local* contraction and yields local linear convergence under weaker assumptions than Bellman completeness.

The idea of using weighting to stabilize Bellman updates has a long history in off-policy temporal-difference learning (Mahmood et al., 2015; Yu, 2015; Sutton et al., 2016; Hallak et al., 2016; Hallak & Mannor, 2017; Patterson et al., 2022), building on earlier work on weighted Bellman equations (Yu & Bertsekas, 2012; Yu et al., 2018). Most of this work focuses on finite-state problems or linear function approximation, although some recent papers consider nonlinear settings (Jiang et al., 2021; Patterson et al., 2022). These ideas are closely related to ours: both use weighting to make off-policy Bellman updates more stable under distribution shift. Emphatic weighting was developed mainly for off-policy evaluation in TD learning, and later extended to control through actor-critic and policy-gradient methods (Imani et al., 2018; Zhang et al., 2019; Stanković et al., 2022; Graves et al., 2023). Patterson et al. (2022) also propose an emphatically weighted, gradient-based TD method for control, though its convergence theory remains open. In contrast, we study weighting in fitted Q-iteration, where the weights modify the regression step rather than a TD recursion, with a focus on nonlinear function approximation. Our analysis also differs: we prove a local contraction result for projected soft Bellman iteration, whereas prior analyses of emphatic weighting typically rely on global contraction arguments. More broadly, our local analysis may also be useful for TD methods in settings where global contraction is hard to establish.

Recent offline RL methods also address instability through pessimistic, minimax, or dual formulations. Pessimistic value-based methods (Jiang et al., 2017; Zanette et al., 2020; Jin et al., 2021; Xie et al., 2021) optimize lower-confidence value estimates, while minimax soft-Q methods (Uehara et al., 2020; Jin et al., 2021; Xie & Jiang, 2021; Uehara et al., 2023) introduce adversarial critics to enforce Bellman constraints under dual realizability and partial coverage. Related entropy-regularized control methods (Ziebart et al., 2008; Haarnoja et al., 2018; Geist et al., 2019; Neu et al., 2017; Zhan et al., 2023; Lan, 2023) optimize policies directly rather than through regression-based fixed-point iteration. Our method instead preserves the fitted value-iteration template and restores contraction through stationary reweighting, without dual critics or pessimistic objectives.

A related line of work studies density-ratio methods for distribution shift by estimating stationary or discounted

occupancy ratios, including classical estimators (Glynn & Henderson, 1998; Paulin, 2015; Meggendorfer, 2023) and DICE-style saddle-point methods (Nachum et al., 2019a; Zhang et al., 2020; Lee et al., 2021; 2022; Nachum et al., 2019b; Wang et al., 2023; Mao et al., 2024). Online variants such as GLOW (Amortila et al., 2024) extend these ideas to adaptively collected data. In this literature, ratios are mainly used to correct for distribution shift; here, we use them to change the regression geometry of fitted Q-iteration.

2. Preliminaries

2.1. Soft Optimal Control

We consider a discounted MDP with continuous state space $\mathcal{S}$, finite action space $\mathcal{A}$, transition kernel P, reward r_0, discount $\gamma \in [0, 1)$, and an arbitrary behavior distribution ν_b over $\mathcal{S} \times \mathcal{A}$. We adopt the entropy-regularized (*soft*) control model used in soft FQI. For any $Q : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ and temperature $\tau > 0$, the softmax policy is $\pi_Q(a \mid s) = \frac{\exp\{Q(s,a)/\tau\}}{\sum_{b\in\mathcal{A}} \exp\{Q(s,b)/\tau\}}$.

Soft evaluation operator. Let $\mathsf{H}(\pi_Q(\cdot \mid S'))$ denote the Shannon entropy of π_Q at S'. Define the soft reward

$$\tilde{r}_Q(s, a) := r_0(s, a) + \gamma\, \mathbb{E}_{S'\sim P(\cdot|s,a)}\big[\tau\, \mathsf{H}(\pi_Q(\cdot \mid S'))\big],$$

and the evaluation operators

$$\mathcal{T}_Q^{\text{eval}} f(s, a) := \tilde{r}_Q(s, a) + \gamma\, (P_{\pi_Q}^{\text{eval}} f)(s, a),$$
$$(P_{\pi_Q}^{\text{eval}} f)(s, a) := \mathbb{E}_{S'\sim P(\cdot|s,a),\, A'\sim\pi_Q(\cdot|S')}\big[f(S', A')\big].$$

The **soft Bellman optimality operator** is

$$(\mathcal{T}Q)(s, a) = r_0(s, a) + \gamma\, \mathbb{E}_{S'\sim P(\cdot|s,a)}\left[\tau \log \sum_{a'\in\mathcal{A}} e^{Q(S',a')/\tau}\right].$$

The **soft optimal Q-function** is

$$Q^\star(s, a) := \sup_\pi Q^\pi(s, a),$$

where the supremum is over all policies π, and the soft evaluation Q-function is

$$Q^\pi(s, a) := \mathbb{E}^\pi_{s,a}\left[\sum_{t=0}^{\infty} \gamma^t \left\{r_0(S_t, A_t) + \tau \mathsf{H}(\pi(\cdot \mid S_t))\right\}\right],$$

where $\mathbb{E}^\pi_{s,a}$ denotes expectation conditional on $S_0 = s$ and $A_0 = a$. The corresponding **soft optimal policy** is denoted by $\pi^\star := \pi_{Q^\star}$, and $Q^\star$ is equivalently the unique bounded fixed point of $\mathcal{T}$, that is, $Q^\star = \mathcal{T}Q^\star$.

Stationary measure and norm. Let $\mu^\star := \mu_{\pi^\star}$ denote the stationary state–action distribution of the chain induced by $(\pi^\star, P)$, that is, $\mu^\star = \mu^\star\pi^\star P$. We measure errors in the Hilbert space $L^2(\mu^\star)$, consisting of measurable functions f such that $\mathbb{E}_{\mu^\star}[f(S, A)^2] < \infty$, equipped with norm

$$\|f\|_{2,\mu^\star} := \big(\mathbb{E}_{\mu^\star}[f(S, A)^2]\big)^{1/2}.$$

Under this norm, $\mathcal{T}_{Q^\star}^{\text{eval}}$ is a γ-contraction (Bertsekas & Tsitsiklis, 1996; Hallak et al., 2016; Patterson et al., 2022). For any nonnegative weight function d, such as the stationary density ratio $\mathrm{d}\mu^\star/\mathrm{d}\nu_b$, let $d \cdot \nu_b$ denote the weighted behavior distribution with Radon–Nikodym derivative d with respect to ν_b.

2.2. Classical FQI and stationary reweighting

Classical dynamic programming computes the optimal Q-function by iterating the Bellman operator $\mathcal{T}$, yielding the fixed-point equation $Q^\star = \mathcal{T}Q^\star$ (Bellman, 1966). With function approximation, this becomes projected Bellman iteration: one applies $\mathcal{T}$ and then projects back onto a function class $\mathcal{F}$. Standard fitted Q-iteration implements this idea using projection under the sampling or behavior distribution. Under Bellman completeness, this projection step is benign, so contraction is preserved. Outside this regime, however, the projection geometry can be misaligned with the Bellman operator, and convergence may fail (Chen & Jiang, 2019; Patterson et al., 2022; Meyn, 2024).

To address this mismatch, we study the population-level projected iteration

$$Q^{(k+1)} = \arg\min_{f\in\mathcal{F}} \|\mathcal{T}(Q^{(k)}) - f\|_{2,\mu^\star}, \tag{1}$$

where $\mu^\star$ is the stationary distribution of the soft-optimal policy $\pi^\star$. Equivalently,

$$Q^{(k+1)} = \mathcal{T}_{\mathcal{F}}(Q^{(k)}), \qquad \mathcal{T}_{\mathcal{F}}(Q) := \Pi_{\mathcal{F}}\mathcal{T}(Q),$$

with $\Pi_{\mathcal{F}}$ the $L^2(\mu^\star)$-projection onto $\overline{\mathcal{F}}$. This iteration is an idealization, since it depends on $\mu^\star$, the stationary distribution of the unknown optimal policy $\pi^\star$. In Section 3.4, we show how to approximate it using the stationary distribution induced by the current iterate, leading to a practical reweighting scheme.

Our key observation is that projection in the stationary norm is aligned with the local contraction geometry of $\mathcal{T}$ around $Q^\star$. Since $\Pi_{\mathcal{F}}$ is nonexpansive in $\|\cdot\|_{2,\mu^\star}$, the projected operator $\mathcal{T}_{\mathcal{F}}$ inherits local contractivity in the same norm. This is the basic reason stationary weighting can recover local convergence guarantees for fitted value iteration beyond Bellman completeness. The next section makes this precise.

2.3. Local Approximation of the Soft Bellman Operator

The next step is to characterize the local behavior of $\mathcal{T}$ near $Q^\star$. The following lemma shows that $\mathcal{T}$ is locally

approximated by the Bellman evaluation operator $\mathcal{T}^{\text{eval}}_{Q^\star}$ associated with the soft-optimal policy $\pi^\star$:

$$\mathcal{T}(Q) \;\approx\; Q^\star + \gamma\, P^{\text{eval}}_{\pi^\star}(Q - Q^\star) \;=\; \mathcal{T}^{\text{eval}}_{Q^\star}(Q), \qquad (2)$$

using the fixed-point identity $Q^\star = \tilde{r}_{Q^\star} + \gamma\, P^{\text{eval}}_{\pi^\star} Q^\star$. Since $P^{\text{eval}}_{\pi^\star}$ is nonexpansive in the stationary $L^2(\mu^\star)$ norm, the evaluation operator $\mathcal{T}^{\text{eval}}_{Q^\star}$ is a γ-contraction (Patterson et al., 2022).

Lemma 1 (First derivative of the soft Bellman operator)**.** *For any Q and perturbation H, the Fréchet derivative of $\mathcal{T}$ at Q is*

$$D\mathcal{T}(Q)[H] \;=\; \gamma\, P^{\text{eval}}_{\pi_Q} H.$$

In particular, at $Q^\star$, $\|D\mathcal{T}(Q^\star)[H]\|_{2,\mu^\star} \;\le\; \gamma\, \|H\|_{2,\mu^\star}$.

The next lemma shows that the linear approximation in (2) is valid when $\mathcal{F} \subseteq L^2(\mu^\star)$ is convex and suitably regular. In the next section, we use this to show that $\mathcal{T}$ locally inherits the contraction property of the evaluation operator.

C1) Stationary action positivity. There exists $\pi_{\min} \in (0,1)$ such that $\pi^\star(A \mid S) \ge \pi_{\min}$ holds $\mu^\star$-almost surely.

C2) Model class convexity. $\mathcal{F} \subseteq L^2(\mu^\star)$ is convex.

C3) Hölder-type L^∞–L^2 relation. There exist $\alpha \in (0,1]$ and $C_\infty < \infty$ such that for all $Q_1, Q_2 \in \mathcal{F} \cup \mathcal{T}(\mathcal{F}) \cup \{Q^\star\}$, $\|Q_1 - Q_2\|_\infty \;\le\; C_\infty\, \|Q_1 - Q_2\|^{\alpha}_{2,\mu^\star}$.

Lemma 2 (Approximation error of linearization)**.** *Under C1–C3, for all $Q \in \mathcal{F}$, define $\beta_{\text{loc}} := \frac{\gamma}{2\tau} C_\infty \sqrt{\frac{|\mathcal{A}|}{\pi_{\min}}}$. Then*

$$\left\|\mathcal{T}(Q) - \mathcal{T}^{\text{eval}}_{Q^\star}(Q)\right\|_{2,\mu^\star} \;\le\; \frac{\beta_{\text{loc}}}{2}\, \|Q - Q^\star\|^{1+\alpha}_{2,\mu^\star}.$$

Condition C1 ensures that $\pi^\star$ assigns each action uniformly positive probability, equivalently, that $Q^\star$ is uniformly bounded. Condition C2 is standard in the analysis, holds for linear classes, and ensures that the projection exists and is nonexpansive (Patterson et al., 2022).

Condition C3 is a mild local Hölder-type smoothness condition controlling the local curvature of $\mathcal{T}$ over $\mathcal{F}$. The associated norm relation holds for many standard function classes. For finite-dimensional linear models of dimension p, it holds with $C_\infty \asymp \sqrt{p}$ and $\alpha = 1$. For reproducing kernel Hilbert spaces with eigenvalue decay $\lambda_j \asymp j^{-2r}$, one may take $\alpha = \frac{2r}{2r+1}$ (Mendelson & Neeman, 2010, Lemma 5.1). It also holds for signed convex hulls of suitable bases (van de Geer, 2014, Lemma 2), and for d-variate Hölder and Sobolev classes of order $s > d/2$ on bounded domains, where $\alpha = 1 - \frac{d}{2s}$ (Bibaut et al. 2021, Lemma 4; Adams & Fournier 2003; Triebel 2006).

3. Local Convergence of Soft Q-Iteration

3.1. Projected soft Q-iteration and local contraction

We study the population projected soft Q-iteration in (1) and establish local contraction and convergence properties of $\mathcal{T}_{\mathcal{F}}$. The stationary reweighting used to restore contraction in standard FQI is introduced in Section 3.4, and the practical fitted algorithm is given in Section 4.

We introduce the following geometric quantities. The minimum $L^2(\mu^\star)$-distance from $Q^\star$ to $\mathcal{F}$ is

$$\text{dist}(\mathcal{F}, Q^\star) := \|\Pi_{\mathcal{F}} Q^\star - Q^\star\|_{2,\mu^\star}.$$

For any $r > 0$ and $Q \in L^2(\mu^\star)$, define

$$\mathbb{B}_{\mathcal{F}}(r, Q) := \{\, f \in \mathcal{F} : \|f - Q\|_{2,\mu^\star} \le r \,\}.$$

We define the local contraction radius by

$$r_0 := \left(\frac{1-\gamma}{\beta_{\text{loc}}}\right)^{1/\alpha}, \qquad \beta_{\text{loc}} := \frac{\gamma}{2\tau}\, C_\infty \sqrt{\frac{|\mathcal{A}|}{\pi_{\min}}}.$$

The corresponding *contraction region* of $\mathcal{T}$ is $\mathbb{B}_{\mathcal{F}}(r_0, Q^\star)$.

Theorem 1 (Local contraction of the soft Bellman operator)**.** *Assume C1–C3 hold. For any $r^\star \in [\text{dist}(\mathcal{F}, Q^\star),\, r_0)$, define*

$$\rho(r^\star) := \gamma + \beta_{\text{loc}}\, (r^\star)^{\alpha}.$$

Then, for all $Q_1, Q_2 \in \mathbb{B}_{\mathcal{F}}(r^\star, Q^\star) \cup \{Q^\star\}$,

$$\|\mathcal{T}(Q_1) - \mathcal{T}(Q_2)\|_{2,\mu^\star} \;\le\; \rho(r^\star)\, \|Q_1 - Q_2\|_{2,\mu^\star}.$$

In particular, the projected operator $\mathcal{T}_{\mathcal{F}} := \Pi_{\mathcal{F}} \mathcal{T}$ is also a $\rho(r^\star)$-contraction on $\mathbb{B}_{\mathcal{F}}(r^\star, Q^\star)$ in $\|\cdot\|_{2,\mu^\star}$.

Banach's fixed-point theorem implies that $\mathcal{T}_{\mathcal{F}}$ admits a locally unique fixed point in a neighborhood of $Q^\star$:

$$Q^\star_{\mathcal{F}} \in \mathcal{F} \quad \text{such that} \quad Q^\star_{\mathcal{F}} = \mathcal{T}_{\mathcal{F}}(Q^\star_{\mathcal{F}}),$$

whenever the gap $\text{dist}(\mathcal{F}, Q^\star)$ is sufficiently small.

Lemma 3 (Existence and local uniqueness of $Q^\star_{\mathcal{F}}$)**.** *Under C1–C3, suppose there exists $r^\star \in (0, r_0)$ such that*

$$\text{dist}(\mathcal{F}, Q^\star) \;\le\; (1 - \rho(r^\star))\, r^\star. \qquad (3)$$

Then $\mathcal{T}_{\mathcal{F}}$ maps $\mathbb{B}_{\mathcal{F}}(r^\star, Q^\star)$ into itself and admits a unique fixed point $Q^\star_{\mathcal{F}}$ in this ball.

3.2. Local linear convergence of soft Q-iteration

We now show that local contraction of $\mathcal{T}_{\mathcal{F}}$ implies linear convergence of soft Q-iteration in a neighborhood of its fixed point. The contraction of $\mathcal{T}_{\mathcal{F}}$ depends on how far its fixed point $Q^\star_{\mathcal{F}}$ is from the true fixed point $Q^\star$, around which $\mathcal{T}$ contracts. We measure this approximation error by

$$\varepsilon_{\mathcal{F}} := \|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu^\star}.$$

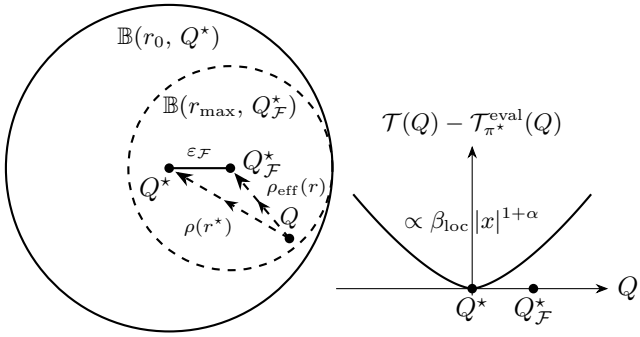

Figure 1. Contraction geometry. Left: contraction region $\mathbb{B}(r_0, Q^\star)$ and basin of attraction $\mathbb{B}(r_{\max}, Q^\star_{\mathcal{F}})$. Right: one–dimensional curvature slice of $\mathcal{T}(Q) - \mathcal{T}^{\text{eval}}_{\pi^\star}(Q)$ near $Q^\star$, with distances $r^\star = \text{dist}(Q, Q^\star)$ and $r = \text{dist}(Q, Q^\star_{\mathcal{F}})$.

We then define the *basin of attraction* as

$$\mathbb{B}_{\mathcal{F}}(r_{\max}, Q^\star_{\mathcal{F}}), \qquad r_{\max} := r_0 - \varepsilon_{\mathcal{F}}.$$

This is the largest ball centered at $Q^\star_{\mathcal{F}}$ contained in the contraction region $\mathbb{B}(r_0, Q^\star)$.

Figure 1 provides some intuition. Inside $\mathbb{B}(r_0, Q^\star)$, the Bellman map $\mathcal{T}$ pulls iterates toward $Q^\star$, while the projection $\Pi_{\mathcal{F}}$ keeps them on $\mathcal{F}$. Starting in $\mathbb{B}_{\mathcal{F}}(r_{\max}, Q^\star_{\mathcal{F}})$, the iterates remain in the contraction region and move toward the constrained fixed point $Q^\star_{\mathcal{F}}$.

We now state the local convergence guarantee. To ensure that the basin of attraction is nonempty, we impose a mild approximate-realizability condition.

C4) Approximate realizability.

$$\text{dist}(\mathcal{F}, Q^\star) \vee \varepsilon_{\mathcal{F}} \;\le\; \frac{\alpha}{\alpha+1}(1-\gamma)\left(\frac{1-\gamma}{(\alpha+1)\,\beta_{\text{loc}}}\right)^{1/\alpha}.$$

By Lemma 3, the bound on $\text{dist}(\mathcal{F}, Q^\star)$ ensures that $Q^\star_{\mathcal{F}}$ exists, so $\varepsilon_{\mathcal{F}}$ is well defined. Since $\text{dist}(\mathcal{F}, Q^\star) \le \varepsilon_{\mathcal{F}}$, Condition C4 is slightly stronger than existence of $Q^\star_{\mathcal{F}}$ alone.

Theorem 2 (Local linear convergence around the projected soft optimum)**.** *Assume C1–C4 hold. Let $r > 0$ satisfy*

$$\varepsilon_{\mathcal{F}} \;\le\; r < r_{\max}, \qquad \rho_{\text{eff}}(r) := \gamma + \beta_{\text{loc}}\,(r + \varepsilon_{\mathcal{F}})^{\alpha}.$$

Then, for any initialization $Q^{(0)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$ and all $k \ge 0$,

$$\|Q^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \;\le\; \rho_{\text{eff}}(r)^k\,\|Q^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star}.$$

The theorem shows that soft Q-iteration is locally contractive around $Q^\star_{\mathcal{F}}$: within any admissible ball $\mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$ with $r < r_{\max}$, each iterate moves toward $Q^\star_{\mathcal{F}}$ at rate $\rho_{\text{eff}}(r)$. By the triangle inequality,

$$\|Q^{(k)} - Q^\star\|_{2,\mu^\star} \;\le\; \|Q^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \;+\; \varepsilon_{\mathcal{F}},$$

so this also yields an error bound relative to $Q^\star$, up to the approximation error $\varepsilon_{\mathcal{F}}$. The result does not cover the infinite-action limit ($|\mathcal{A}| \uparrow \infty$), the hard-max limit ($\tau \downarrow 0$), or the undiscounted limit ($\gamma \to 1$) without additional assumptions, since the contraction weakens and the basin of attraction vanishes in these regimes.

3.3. Contraction rates and radius dependence

Theorem 2 is intentionally conservative: the modulus $\rho_{\text{eff}}(r)$ depends on the initialization radius r and gives a uniform bound over all later iterations. As the iterates approach $Q^\star_{\mathcal{F}}$, they enter smaller basins where contraction is stronger. In particular, if $Q^{(K)} \in \mathbb{B}_{\mathcal{F}}(r', Q^\star_{\mathcal{F}})$ with $r' < r$, we may reapply Theorem 2 with initialization $Q^{(0)} := Q^{(K)}$, yielding the sharper modulus $\rho_{\text{eff}}(r')$. Along the trajectory, $\rho_{\text{eff}}(r)$ decreases toward $\rho_{\min} := \gamma + \beta_{\text{loc}}\varepsilon^{\alpha}_{\mathcal{F}}$. Under misspecification, this is the smallest attainable modulus, and it approaches γ only when $\varepsilon_{\mathcal{F}}$ is small.

The following corollary shows that, in the realizable case, the contraction modulus approaches γ at a geometric rate.

Corollary 1 (Finite-time tightening of the contraction rate)**.** *Assume $Q^\star_{\mathcal{F}} = Q^\star$ and the conditions of Theorem 2. Define $e_k := \|Q^{(k)} - Q^\star\|_{2,\mu^\star}$ and $\rho := \gamma + \beta_{\text{loc}} r^{\alpha}$. Then the per-iteration contraction modulus satisfies*

$$\rho_k := \frac{e_{k+1}}{e_k} \le \gamma + \beta_{\text{loc}} e_k^{\alpha},$$

and $\rho_k - \gamma \le \beta_{\text{loc}} e_0^{\alpha} \rho^{\alpha k}, \quad k \ge 0.$

The exponent α in Condition C3 governs how quickly the contraction modulus improves along the trajectory. Larger values of α yield faster improvement. When $\alpha = 1$, the remainder is quadratic, and $\rho_k - \gamma \le \beta_{\text{loc}} e_0 \rho^k$.

3.4. Restoring contraction via stationary reweighting

The population recursion studied thus far is an idealization, since the projection $\Pi_{\mathcal{F}}$ is defined with respect to $\mu^\star$, the stationary distribution of the unknown optimal policy $\pi^\star$. Our proposal is to approximate this update by reweighting each regression step under the behavior distribution using the stationary density ratio of the current policy.

Admissible weighting functions. Our theory does not require the exact stationary density ratio $\mathrm{d}\mu^\star/\mathrm{d}\nu_b$, but only a weight d that induces the same projection geometry as the target stationary distribution $\mu^\star$. This is weaker than Bellman completeness ($\mathcal{T}(\mathcal{F}) \subseteq \mathcal{F}$), under which the choice of weight is irrelevant.

C5) Bellman residual orthogonality under weighting. There exists $d \in L^2(\nu_b)$ such that for all $f, Q \in \mathcal{F}$,

$$\mathbb{E}_{\nu_b}\Big[d(S,A)\, f(S,A)\, \big(\mathcal{T}Q - \Pi_{\mathcal{F}}\mathcal{T}Q\big)(S,A)\Big] = 0.$$

Condition C5 requires that the Bellman residuals be orthogonal to $\mathcal{F}$ under the weighted inner product on $L^2(d \cdot \nu_b)$, for all $Q \in \mathcal{F}$. It ensures that the projected Bellman update

$$Q^{(k+1)} = \mathcal{T}_{\mathcal{F}}(Q^{(k)}) = \arg\min_{f \in \mathcal{F}} \|\mathcal{T}(Q^{(k)}) - f\|_{2,\mu^\star}$$

can equivalently be written as the weighted regression

$$Q^{(k+1)} = \arg\min_{f \in \mathcal{F}} \mathbb{E}_{\nu_b}\Big[d(S,A)\{\mathcal{T}(Q^{(k)})(S,A) - f(S,A)\}^2\Big]$$

Under Bellman completeness, the orthogonality condition holds automatically, since the Bellman residual is zero.

Population iteration. We now present the population analogue of our practical algorithm. Initialize $Q^{(0)} \in \mathcal{F}$ in the basin of attraction $\mathbb{B}(r_{\max}, Q^\star_{\mathcal{F}})$. For $k = 0, \dots, K$, let $d^{(k)} \propto \mathrm{d}\mu_{\pi_{Q^{(k)}}}/\mathrm{d}\nu_b$ be the stationary ratio of the policy induced by $Q^{(k)}$, and update

$$Q^{(k+1)} := \arg\min_{Q \in \mathcal{F}} \mathbb{E}_{\nu_b}\Big[d^{(k)}(S,A)\,\{(\mathcal{T}Q^{(k)} - Q)(S,A)\}^2\Big].$$

This is standard projected soft Q-iteration, except that each regression step is reweighted to match the stationary geometry of the policy induced by the current iterate.

4. Stationary-Reweighted Soft FQI

We now turn from the population operator to a practical finite-sample algorithm. In *stationary-reweighted soft FQI*, the stationary weights, Bellman targets, and regression updates are all estimated from offline off-policy data.

Algorithm. Let $\mathcal{D}_n := \{(S_i, A_i, R_i, S_i')\}_{i=1}^n$ denote a batch of transitions collected under the behavior distribution ν_b, and let $\mathcal{F}$ be the chosen function class. Initialize $\widehat{Q}^{(0)}$, form the induced softmax policy $\pi_{\widehat{Q}^{(0)}}$, and estimate initial weights $\widehat{d}^{(0)}$ by approximating the stationary ratio $\mathrm{d}\mu_{\pi_{\widehat{Q}^{(0)}}}/\mathrm{d}\nu_b$ (or, more generally, any weight satisfying Condition C5).

For each $k \in \{0, \dots, K-1\}$, form the empirical soft Bellman targets

$$\widehat{y}_i^{(k)} = R_i + \gamma\tau \log \sum_{a'} \exp\big(\widehat{Q}^{(k)}(S_i', a')/\tau\big).$$

Then update the Q-estimate via weighted least squares:

$$\widehat{Q}^{(k+1)} \in \arg\min_{Q \in \mathcal{F}} \frac{1}{n}\sum_{i=1}^n \widehat{d}^{(k)}(S_i, A_i)\,\big(\widehat{y}_i^{(k)} - Q(S_i, A_i)\big)^2.$$

Finally, re-estimate an admissible stationary ratio for the induced policy $\pi_{\widehat{Q}^{(k+1)}}$ to obtain updated weights $\widehat{d}^{(k+1)}$.

Estimating stationary ratios. The stationary density ratio $\mathrm{d}\mu_\pi/\mathrm{d}\nu_b$ can be estimated using stationarity-matching saddle-point objectives from the DICE family (Nachum et al., 2019a; Zhang et al., 2020; Lee et al., 2021; 2022), minimax methods (Uehara et al., 2020), or approximate balancing weights (Wang et al., 2023); see Appendix A for details. In our setting, these ratios are used only to learn a stationary weighting function d, not to construct an unbiased value estimator. Since the weights affect only the regression geometry, even coarse or misspecified estimates, including those from simple linear models, may still improve the stability of FQI when Bellman completeness fails, echoing the motivation behind emphatic weighting (Patterson et al., 2022).

Initialization and weight refreshing. Our local guarantees require the iterates to enter the basin of attraction $\mathbb{B}_{\mathcal{F}}(r_{\max}, Q^\star_{\mathcal{F}})$. In practice, a short warm start—for example, a few iterations of unweighted soft FQI or another preliminary estimator—together with temperature annealing may bring the initialization closer to the fixed point. To reduce computational cost, one may also refresh the weights every few iterations rather than at every iteration.

Towards global convergence via temperature annealing. To ensure initialization within the basin of attraction, one can start at a large temperature τ so that the softmax policy is essentially uniform, where the operator is smoother and the contraction radius scales as $r_0 \asymp \tau^{1/\alpha}$. This motivates a *temperature homotopy*: warm-start at high τ, run stationary-reweighted updates to convergence, then gradually decrease τ, re-solving at each stage while staying within that stage's contraction region. This continuation perspective parallels classical homotopy methods in nonlinear optimization (e.g., Allgower & Georg, 1990; Kelley, 2003; Nesterov, 2005). In the low-temperature regime ($\tau \in (0, 1]$), preliminary analysis (Appendix B) suggests validity under additional margin and action-gap conditions (Uehara et al., 2023), which keep the contraction radius bounded as $\tau \downarrow 0$. Developing a practical version of this scheme and its theory is left to future work.

Sources of error. The algorithm incurs three sources of approximation: (i) sampling error; (ii) function-approximation error; and (iii) weight estimation error. All three components are reflected in the finite-sample analysis below.

5. Finite-Sample Convergence Analysis

5.1. Inexact local Picard iteration

Our finite-sample analysis departs from classical global arguments for fitted value iteration (Munos & Szepesvári, 2008). Instead, we develop a *local* analysis that accounts for local contractivity of the Bellman operator and iterative weight estimation.

We begin with an inexact-iteration lemma, a local analogue

of the classical recursion of Munos & Szepesvári (2008).

Lemma 4 (Inexact iteration error bound)**.** *Assume C1-C5. Fix any $0 < r < r_{\max}$ and let $\rho_\star := \rho_{\mathrm{eff}}(r)$ be the local contraction modulus from Theorem 2. For $k \geq 0$, consider inexact updates*

$$\widehat{Q}^{(k+1)} = \mathcal{T}_{\mathcal{F}}\Big(\widehat{Q}^{(k)}\Big) + \vec{\xi}_k, \qquad \|\vec{\xi}_k\|_{2,\mu^\star} \leq \xi_k.$$

Assume $\|\widehat{Q}^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \leq r$ and $\xi_k \leq (1-\rho_\star)\, r$ for all $k \geq 0$. Then, for every $k \geq 0$,

$$\|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \leq \rho_\star^k \, \|\widehat{Q}^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} + \sum_{j=0}^{k-1} \rho_\star^{k-1-j}\, \xi_j.$$

The lemma assumes that $\widehat{Q}^{(0)}$ is initialized in $\mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$, and the condition $\xi_k \leq (1-\rho_\star)r$ ensures that the iterates remain in this basin, where $\mathcal{T}_{\mathcal{F}}$ is $\rho_\star$-contractive. Lemma 4 then yields finite-sample linear convergence up to the error floor $\sum_{j=0}^{k-1} \rho_\star^{k-1-j}\xi_j$. It therefore remains to control the per-iteration inexactness term $\|\widehat{Q}^{(k+1)} - \mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)})\|_{2,\mu^\star}$ with high probability, accounting for sampling noise, approximation error, and weight-estimation error.

5.2. Finite-Sample Bounds on the Per-Iteration Error

We now derive a high-probability bound for this per-iteration error in terms of the statistical complexity of the function class used at each step. Define the extended class

$$\mathcal{F}_{\mathrm{ext}} := \mathcal{F} \;\cup\; \{\pi_Q f : f \in \mathcal{F},\; Q \in \mathcal{F}\},$$

which captures both elements of $\mathcal{F}$ and their soft–Bellman-evaluated counterparts under policies induced by $\mathcal{F}$.

For $\varepsilon > 0$, let $N(\varepsilon, \mathcal{F}_{\mathrm{ext}}, L^2(Q))$ denote the smallest number of $L^2(Q)$-balls of radius ε needed to cover $\mathcal{F}_{\mathrm{ext}}$. Define the uniform entropy integral (Van Der Vaart & Wellner, 1996)

$$\mathcal{J}(\delta, \mathcal{F}_{\mathrm{ext}}) := \int_0^\delta \sup_Q \sqrt{\log N(\varepsilon, \mathcal{F}_{\mathrm{ext}}, L^2(Q))}\, d\varepsilon,$$

and let the associated critical radius (Bartlett et al., 2005) be

$$\delta_n := \inf\Big\{\delta > 0 : \frac{\mathcal{J}(\delta, \mathcal{F}_{\mathrm{ext}})}{\sqrt{n}\,\delta^2} \leq 1\Big\}.$$

C6) Stationary–weighted norm equivalence. There exist $0 < c_0 \leq c_1 < \infty$ such that for all $f \in L^2(\mu^\star)$,

$$c_0\|f\|_{2,\mu^\star} \;\leq\; \|f\|_{2,\, d\cdot\nu_b} \;\leq\; c_1\|f\|_{2,\mu^\star}.$$

C7) Uniform boundedness. The functions d, $1/d$, the reward R, and the iterates $\{\widehat{d}^{(k)}, \widehat{Q}^{(k)}, \mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)})\}$ are uniformly bounded by a constant $M < \infty$.

C8) Sample splitting. The weight estimators $\{\widehat{d}^{(k)}\}_{k=0}^K$ are computed using data independent of the batch $\mathcal{D}_n$ used for the regression updates.

C9) Complexity control. Both $\mathcal{J}(\infty, \mathcal{F}_{\mathrm{ext}}) < \infty$ and

$$\frac{\mathcal{J}(\delta, \mathcal{F}_{\mathrm{ext}})}{\delta\sqrt{\mathrm{loglog}(1/\delta)}} \;\to\; \infty \qquad \text{as } \delta \to 0.$$

Condition C6 translates weighted–behavior errors into stationary-norm guarantees and is automatically satisfied when d is the stationary density ratio. C7 is a standard simplification and can be relaxed via truncation and tail conditions. C8 ensures independence between weight estimation and regression and can be removed via cross-fitting (Nie & Wager, 2021; Foster & Syrgkanis, 2023). Finally, C9 controls localized Rademacher complexities, following Munos & Szepesvári (2008).

Fix a horizon $K \geq 1$ and confidence level $\eta \in (0,1)$, and define the statistical error

$$\delta_{\mathrm{stat}}(n, \eta, K) := \delta_n + \sqrt{\frac{\log(K/\eta)}{n}},$$

and the weight error

$$\varepsilon_{\mathrm{wt}}(k) := \big\|\widehat{d}^{(k)}/d - 1\big\|_{2,\mu^\star}.$$

We further define the local Bellman approximation error

$$\begin{aligned}\varepsilon_{\mathrm{Bell}}(k) &:= \big\|\mathcal{T}(\widehat{Q}^{(k)}) - \mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)})\big\|_{2,\mu^\star} \\ &\leq\; \varepsilon_{\mathcal{F}} \;+\; 2\,\big\|\mathcal{T}(\widehat{Q}^{(k)}) - \mathcal{T}(Q^\star)\big\|_{2,\mu^\star}.\end{aligned} \tag{4}$$

This quantity vanishes under Bellman completeness, and is small whenever $\widehat{Q}^{(k)}$ and $Q^\star_{\mathcal{F}}$ lie close to $Q^\star$.

Lemma 5 (Statistical accuracy of one weighted regression)**.** *Assume C3,C5–C9. There exists $C > 0$ such that, for all $\eta \in (0,1)$, $0 \leq k \leq K-1$, with probability at least $1-\eta$,*

$$\begin{aligned}\big\|\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\big\|_{2,\mu^\star} \;&\leq C\delta_{\mathrm{stat}}(n,\eta,K) \\ &+\; C\left(\varepsilon_{\mathrm{wt}}(k)^{\frac{1}{1-\alpha}} \;+\; \varepsilon_{\mathrm{wt}}(k)\,\varepsilon_{\mathrm{Bell}}(k)\right).\end{aligned}$$

The first term, $\delta_{\mathrm{stat}}(n,\eta,K)$, is the usual statistical error from empirical risk minimization over $\mathcal{F}$. Provided $\varepsilon_{\mathrm{wt}}(k) < 1$, the leading weight-error term is $\varepsilon_{\mathrm{wt}}(k)$ as $\alpha \to 0$, whereas as $\alpha \to 1$ it becomes $\varepsilon_{\mathrm{Bell}}(k)\,\varepsilon_{\mathrm{wt}}(k)$. Thus, the effect of weight misspecification is attenuated when the Bellman approximation error is small. In particular, approximate Bellman completeness can mitigate weight-estimation error when $\mathcal{F}$ is sufficiently regular, that is, for larger α.

For s–sparse linear models in dimension d, $\delta_n \asymp \sqrt{s\log(d/s)/n}$ (Wainwright, 2019). For VC-subgraph classes of dimension V, $\delta_n \asymp \sqrt{V\log(n/V)/n}$ (Van Der Vaart & Wellner, 1996). For Hölder or Sobolev classes of smoothness s with $d \le 2s$, $\delta_n \asymp n^{-s/(2s+d)}$ (Nickl & Pötscher, 2007). For finite Hardy–Krause variation classes, $\delta_n \asymp n^{-1/3}(\log n)^{2(d-1)/3}$ (Bibaut & van der Laan, 2019).

5.3. Finite-Sample Convergence Bound (main result)

With high-probability bounds for the per-iteration errors in hand, we combine Lemma 4 and Lemma 5 to obtain our main finite-sample convergence bound.

We define the per–iteration inexactness

$$\xi_k := C\Big(\delta_{\mathrm{stat}}(n,\eta,K) + \varepsilon_{\mathrm{wt}}(k)^{\frac{1}{1-\alpha}} + \varepsilon_{\mathrm{wt}}(k)\,\varepsilon_{\mathcal{F}}\Big),$$

where C is the constant in Lemma 5. We introduce the weight–adjusted contraction factor

$$\rho_{\mathrm{eff,wt}}(r,k) := \rho_{\mathrm{eff}}(r)\,\big(1 + 2C\,\varepsilon_{\mathrm{wt}}(k)\big),$$

and its worst–case version

$$\rho_\star(r) := \rho_{\mathrm{eff}}(r)\,\big(1{+}2C\,\bar{\varepsilon}_{\mathrm{wt}}\big), \qquad \bar{\varepsilon}_{\mathrm{wt}} := \max_{0\le j\le K-1} \varepsilon_{\mathrm{wt}}(j).$$

C10) Basin stability. There exists $r \in (0, r_{\max})$ such that, for all $0 \le k \le K-1$,

$$\rho_{\mathrm{eff,wt}}(r,k) < 1, \qquad \xi_k \;\le\; \big(1 - \rho_{\mathrm{eff,wt}}(r,k)\big)\,r.$$

Condition C10 ensures that there exists a ball $\mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$ on which the ideal update is contractive and the per-iteration inexactness is small enough to keep the iterates inside this ball. The bound $\xi_k \le (1-\rho_{\mathrm{eff,wt}}(r,k))\,r$ ensures that, after the contractive step pulls the iterate to radius $\rho_{\mathrm{eff,wt}}(r,k)\,r$, the remaining additive error does not push it outside $\mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$. As ξ_k decreases (e.g., with larger sample sizes), it becomes easier to choose $r \in (0, r_{\max})$ so that the condition holds.

Our main convergence theorem is stated below. We quantify the cumulative weight–estimation error by

$$\Delta_{\mathrm{wt}}(k,\alpha) := (1-\rho_\star)\sum_{j=0}^{k-1} \rho_\star^{k-1-j}\,\varepsilon_{\mathrm{wt}}(j)^{\frac{1}{1-\alpha}}.$$

Theorem 3 (Local convergence with geometrically damped weight error)**.** *Assume Conditions C1–C10 hold for some $r \in (0, r_{\max})$ with $\rho_\star := \rho_\star(r)$, and that $\widehat{Q}^{(0)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$. Then, there exists $C' < \infty$, with probability at least $1-\eta$, for all $1 \le k \le K$,*

$$\begin{aligned}\big\|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\big\|_{2,\mu^\star} \;\le\; & \rho_\star^k\,\big\|\widehat{Q}^{(0)} - Q^\star_{\mathcal{F}}\big\|_{2,\mu^\star} \\ & + \frac{C'}{1-\rho_\star}\,\delta_{\mathrm{stat}}(n,\eta,K) \\ & + \frac{C'}{1-\rho_\star}\Big(\Delta_{\mathrm{wt}}(k,\alpha) + \varepsilon_{\mathcal{F}}\,\Delta_{\mathrm{wt}}(k,0)\Big).\end{aligned}$$

Structure of the final error bound. Theorem 3 yields an inexact Picard recursion: the bias contracts at rate $\rho_\star$, while the statistical error $\delta_{\mathrm{stat}}(n,\eta,K)$ and the discounted weight errors $\Delta_{\mathrm{wt}}(k,\alpha)$ enter additively (with the usual $(1-\rho_\star)^{-1}$ prefactor for the noncontractive terms). This mirrors the classical sup-norm analysis of Munos & Szepesvári (2008), but now in a locally contractive stationary $L^2(\mu^\star)$ geometry. Updating the weights at each iteration ensures that weight-induced errors are *geometrically damped* through $\Delta_{\mathrm{wt}}(k,\alpha)$; the remaining weight-induced contribution is the approximation term $\varepsilon_{\mathcal{F}}\,\Delta_{\mathrm{wt}}(k,0)$. In contrast with existing FQI analyses, our guarantees do not require Bellman completeness; instead, they rely on weights satisfying C5 together with approximate realizability.

A consequence of Theorem 3 is that the iterate error depends on the weight errors only through the higher-order term $\varepsilon_{\mathrm{wt}}(k)^{1/(1-\alpha)}$ (up to the approximation floor $\varepsilon_{\mathcal{F}}$). Thus, even without refreshing weights each iteration, the iterates contract to a neighborhood whose radius is set by the statistical error, this powered weight-error term, and $\varepsilon_{\mathcal{F}}$.

Geometric error decay under weight refreshing. If the weight errors decay as $\varepsilon_{\mathrm{wt}}(k) = b^k\varepsilon_{\mathrm{wt}}(0)$ with $b \in (0,1)$, then $\Delta_{\mathrm{wt}}(k,\alpha)$ also decays geometrically with base $\min\{b^{1/(1-\alpha)}, \rho_\star\}$ (up to polynomial factors in k). With Theorem 3, this implies near-geometric decay of the iterate error up to the statistical and approximation floors. This discussion assumes *a priori* geometric decay of the weight errors. In practice, they depend on the iterates, creating a feedback loop: contraction of $\widehat{Q}^{(k)}$ requires accurate weights, while accurate weights require staying in the contraction region. The next theorem breaks this loop via a mild polynomial coupling between iterate and weight errors, yielding a joint recursion and guarantees from the initialization error alone.

C11) Weight–error coupling. For any $\eta \in (0,1)$, there exist constants $C' < \infty$ and $\kappa > 0$ such that, with probability at least $1-\eta$, the following holds for all $0 \le k \le K-1$:

$$\varepsilon_{\mathrm{wt}}(k) \;\le\; C'\,\|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|^{\kappa}_{2,\mu^\star} \;+\; \delta_{\mathrm{wt}}(n,\eta,K).$$

Condition C11 decomposes the weight–estimation error at round k into a component controlled by the iterate error and a statistical remainder $\delta_{\mathrm{wt}}(n,\eta,K)$.

We define the r-dependent per–iteration inexactness

$$\xi_\star(r) := \Delta_{\mathrm{stat}} + \Phi(r),$$

where the nonlinear error term is $\Phi(r) := r^{1+\kappa} + r^{\frac{\kappa}{1-\alpha}} + \varepsilon_{\mathcal{F}}\,r^{\kappa}$, and the total statistical error is

$$\Delta_{\mathrm{stat}} := \varepsilon_{\mathrm{stat}}(n,\eta,K) + \delta_{\mathrm{wt}}(n,\eta,K)^{\frac{1}{1-\alpha}} + \varepsilon_{\mathcal{F}}\,\delta_{\mathrm{wt}}(n,\eta,K).$$

For a constant $B < \infty$ determined only by the structural constants in our assumptions, define the contraction factor

$$\bar{\rho}(r) := \rho_{\text{eff}}(r) + B\,\delta_{\text{wt}}(n, \eta, K).$$

C12) Basin stability. There exists $r \in (0, r_{\max})$ such that, for all $0 \le k \le K-1$,

$$\bar{\rho}(r) < 1, \qquad B\,\xi_\star(r) \le (1 - \bar{\rho}(r))\, r.$$

Theorem 4 (Local convergence with decoupled errors)**.** *Assume C1–C9, C11, and C12 hold for some $r \in (0, r_{\max})$ with $\bar{\rho} := \bar{\rho}(r)$, and that $\widehat{Q}^{(0)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$. Then, with probability at least $1-\eta$, the errors $e_k := \|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star}$ satisfy, for some constant $C < \infty$ and all $1 \le k \le K$,*

$$e_k \le \bar{\rho}^{\,k} e_0 \;+\; B' \sum_{j=0}^{k-1} \bar{\rho}^{\,k-1-j}\, \Phi(e_j) \;+\; B'\, \frac{\Delta_{\text{stat}}}{1-\bar{\rho}}.$$

Theorem 4 shows that the iterates contract at the geometric rate $\bar{\rho}^{\,k} e_0$, up to a discounted sum of the nonlinear errors $\Phi(e_j)$. When $\kappa > 1 - \alpha$, all terms in $\Phi(e_j)$ are strictly higher order near the fixed point, and the remaining e_j^κ term is scaled by the approximation error $\varepsilon_{\mathcal{F}}$. In this regime, the sum $\sum_{j=0}^{k-1} \bar{\rho}^{\,k-1-j}\Phi(e_j)$ is negligible relative to $\bar{\rho}^{\,k} e_0$, yielding geometric convergence up to the statistical floor Δ_{stat} and the approximation error $\varepsilon_{\mathcal{F}}$.

6. Illustrative Experiments

We present a controlled tabular experiment intended only to illustrate the mechanism behind our theory. Specifically, we study soft fitted Q-iteration under severe norm mismatch in a setting where $Q^\star_\tau$ is known, so that the effects of stationary reweighting and temperature homotopy can be isolated.

Setup. We generate Garnet MDPs (Bhatnagar et al., 2009) with $|\mathcal{S}| = 50$, $|\mathcal{A}| = 4$, branching factor 5, rewards $R \sim \mathcal{N}(0, 0.1^2)$, and $\gamma = 0.99$. We target the soft-optimal policy at $\tau = 10^{-6}$ (near hard-max). The behavior policy is drawn independently at each state from a uniform Dirichlet distribution, inducing strong mismatch. We use a linear class $\mathcal{F} = \{\phi(s,a)^\top \theta : \theta \in \mathbb{R}^5\}$ that contains $Q^\star_\tau$ but is not Bellman complete, and collect 10^5 reset-sampled one-step transitions. Results are averaged over $M = 200$ random MDPs and datasets. We compare standard soft FQI to stationary-weighted FQI (SW-FQI), both run directly at τ and with temperature homotopy from $\tau_{\text{init}} = 0.1$ to τ (followed by continued training at τ), for 300 regression steps total. We report $\|Q_k - Q^\star_\tau\|^2_{2,\mu^\star}$.

Results. Figure 2 illustrates the predicted qualitative behavior. SW-FQI contracts faster and reaches a lower error than standard FQI at fixed τ, and remains consistently faster along the annealing path. With a very high-temperature start

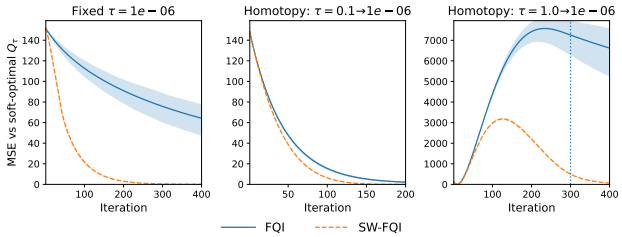

Figure 2. Soft FQI under severe norm mismatch on a Garnet MDP. Left: direct training at τ. Middle/right: temperature annealing. Shaded regions show 25%–75% quantiles across seeds; the dashed line marks when τ reaches its target.

($\tau_{\text{init}} = 1$), standard FQI is initially unstable because the high-τ fixed point is severely misspecified, whereas SW-FQI is only mildly perturbed and re-stabilizes quickly as τ decreases. Overall, temperature continuation improves early conditioning, while stationary norm alignment stabilizes the dynamics throughout.

7. Discussion and Conclusion

We developed a local contraction analysis for fitted soft Q-iteration under stationary reweighting. Our results show that, even without Bellman completeness, the fitted Bellman update can be locally contractive around its fixed point, yielding linear convergence within an explicit basin of attraction. More broadly, this highlights local contraction as a useful lens for approximate dynamic programming under function approximation, especially when global contraction is too strong to expect.

Our analysis also suggests a simple algorithmic principle: when the Bellman operator is naturally contractive in one norm but regression is performed in another, changing the regression geometry can improve stability. In our setting, stationary reweighting reduces the mismatch between the target policy's stationary norm and the behavior distribution underlying fitted updates. This connects fitted value methods to the emphatic-weighting literature in TD learning and suggests that weighting is a useful design principle for fitted value methods more broadly.

Several important directions remain open. Our guarantees are local and therefore depend on initialization, making reliable basin-entry strategies an important next step. Estimating stationary ratios also remains challenging, especially under weak coverage, so practical success will depend on weight estimators that are both stable and computationally tractable. It would also be valuable to extend the analysis beyond the i.i.d. setting to trajectory data with temporal dependence, to better understand the low-temperature limit toward hard-max FQI, and to evaluate empirically when stationary reweighting offers meaningful gains over standard fitted methods, emphatic-style corrections, and minimax or

density-ratio approaches.

Impact Statement.

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Estimation of Stationary Ratios

A key component of stationary-reweighted FQI is estimating the stationary density ratio. This is challenging in its own right and typically leads to a minimax or saddle-point problem, as in DICE-style methods (Nachum et al., 2019a; Zhang et al., 2020; Lee et al., 2021; Wang et al., 2023) and related minimax weight-estimation approaches (Uehara et al., 2020).

Since $d_\pi \nu_b = \mu_\pi$ is a stationary distribution for (π, P), the stationary ratio $d_\pi = \mathrm{d}\mu_\pi / \mathrm{d} \cdot \nu_b$ satisfies the moment condition

$$\mathbb{E}\big[d_\pi(S,A)\,\big(f(S,A) - f(S',A')\big)\big] = 0 \quad \text{for all } f, \tag{5}$$

where $(S,A) \sim \nu_b$, $S' \sim P(\cdot \mid S,A)$, and $A' \sim \pi(\cdot \mid S')$.

Minimax and DICE-style saddle-point methods enforce stationarity through a variational formulation and learn the ratio using paired weight and critic classes. A representative empirical objective is

$$\arg\min_{d \in \mathcal{G}} \sup_{f \in \mathcal{H}} \left\{ \frac{1}{n} \sum_{i=1}^{n} d(S_i, A_i)\,\big(f(S_i, A_i) - f(S_i', A_i')\big) - \frac{1}{2n} \sum_{i=1}^{n} f(S_i, A_i)^2 \right\} + \left(\frac{1}{n} \sum_{i=1}^{n} d(S_i, A_i) - 1 \right)^2,$$

where $A_i' \sim \pi(\cdot \mid S_i')$, and the final term enforces normalization. When the critic is restricted to a linear function class or an RKHS, the inner maximization can often be expressed in closed form (Olivas-Martinez & Rotnitzky, 2025). In practice, such structured critic classes can yield simpler and more stable estimators than highly flexible saddle-point methods, while still producing useful weights; see, for example, approximate balancing weights (Wang et al., 2023). More generally, theoretical analyses of minimax-style weight estimation are given by Dikkala et al. (2020); Uehara et al. (2020); Wang et al. (2023).

Importantly, weighting can help even without accurate stationary-ratio estimation. Because the weights define only the regression norm, even misspecified weights may improve stability by better aligning the update with the on-policy geometry. See Patterson et al. (2022) for further discussion.

B. Towards Global Convergence via Temperature Annealing

B.1. Overview of challenges

Global convergence via temperature annealing. Our local linear convergence guarantees require that the initialization lie sufficiently inside the contraction region. A natural way to ensure this is to begin with a large softmax temperature τ, for which the contraction radius $r_0(\tau)$ grows as $\tau^{1/\alpha}$ while the soft–hard bias scales only as $O(\tau)$. Thus, when $\tau > 1$, the stationary-reweighted update reliably converges to $Q^\star_{\mathcal{F}}(\tau)$ from essentially any reasonable initialization. This suggests a *temperature– annealing* strategy: warm-start at a large temperature, solve the projected Bellman equation, then decrease τ gradually while keeping each intermediate iterate inside its associated local basin. This continuation viewpoint parallels classical homotopy methods for extending local fixed-point analyses (e.g., Allgower & Georg, 1990; Kelley, 2003; Nesterov, 2005).

The main difficulty is that for $\tau < 1$, the baseline curvature bound $\beta_{\mathrm{loc}}(\tau) = O(1/\tau)$ causes the contraction region to shrink faster than the $O(\tau)$ bias, so the continuation path may exit the local basin as $\tau \downarrow 0$. Under a uniform action-gap condition at the hard-max solution, however, the curvature decays exponentially in $1/\tau$, yielding an enhanced radius $r_0^{\mathrm{gap}}(\tau)$ that remains bounded away from zero as $\tau \downarrow 0$. This restores the viability of the homotopy path down to small temperatures.

The next subsection formalizes the structural assumptions under which these refinements hold and sketches how our local theory can, in principle, be extended toward global behavior. A full theoretical and empirical development of temperature– annealing schemes remains an interesting direction for future work.

Remark. For fixed τ, one may instead use a *discount–factor homotopy*, slowly increasing γ so that the soft-optimal fixed point drifts smoothly and the iterates remain within

each stage's local contraction region. Since $r_0(\gamma,\tau) \asymp \big((1-\gamma)/\gamma\big)^{1/\alpha}$ for fixed τ, the step sizes required to stay within the basin shrink rapidly as $\gamma \uparrow 1$, reflecting the collapse of the contraction region at long horizons. In practice, combining temperature and discount–factor homotopies may further improve stability.

B.2. A Path Toward Global Behavior: Assumptions and Heuristics

In this section we index the softmax temperature by τ, writing $Q^\star(\tau)$ for the soft-optimal value function and $Q^\star_{\text{hard}}$ for the hard-optimal limit. Under Conditions C1–C3, our local theory yields $r_0(\tau) \asymp \tau^{1/\alpha}$ and $\|Q^\star(\tau) - Q^\star_{\text{hard}}\|_\infty = O(\tau)$. As $\tau \downarrow 0$, however, the curvature parameter $\beta_{\text{loc}}(\tau)$ grows as $O(1/\tau)$, causing the contraction region to collapse. To extend the analysis to small temperatures, one must prevent this collapse.

A natural structural requirement is a uniform action gap at the hard-max optimum. Under this margin condition (together with a local gap-invariance assumption), the covariance term in the second derivative of the soft Bellman operator decays exponentially in $1/\tau$. This yields a much smaller effective curvature $\beta_{\text{loc}}(\tau)$ and maintains a nonvanishing contraction radius down to $\tau \approx 0$, suggesting a path toward a global analysis via a temperature– annealing argument.

To formalize the improved curvature bound, we introduce two structural assumptions used only in this appendix:

S1 Uniform action gap. There exists $\Delta > 0$ such that, $\mu^\star$-almost surely, for all $a \neq A^\star(S)$,

$$Q^\star_{\text{hard}}(S, A^\star(S)) \;\geq\; Q^\star_{\text{hard}}(S,a) + 2\Delta,$$

for a measurable optimal action $A^\star(S)$.

S2 Local neighborhood around $Q^\star_{\text{hard}}$. Let $r_{\text{gap}} > 0$ satisfy

$$C_\infty\, r_{\text{gap}}^\alpha \;\leq\; \Delta,$$

and define the localized region

$$\mathbb{B}_{\text{gap}} := \big\{\, Q \in \mathcal{F} : \|Q - Q^\star_{\text{hard}}\|_{2,\mu^\star} \leq r_{\text{gap}} \big\}.$$

For any $Q \in \mathbb{B}_{\text{gap}}$, the action $A^\star(S)$ remains uniquely optimal with margin at least Δ.

Under Assumptions S1–S2, Lemma 8 can be substituted for Lemma 7 throughout the local analysis, replacing $\beta_{\text{loc}}(\tau)$ by

$$\beta^{\text{gap}}_{\text{loc}}(\tau) := \frac{\gamma}{\tau}\, C_\infty \sqrt{\frac{|\mathcal{A}|}{\pi_{\min}}}\, C_{\text{gap}}\, e^{-\Delta/(2\tau)}.$$

The admissible contraction radius must then lie inside both the geometric radius and the margin-stable ball $\mathbb{B}_{\text{gap}}$, giving

$$r_0^{\text{gap}}(\tau) := \min\left\{ r_{\text{gap}},\; \left(\frac{1-\gamma}{\beta^{\text{gap}}_{\text{loc}}(\tau)}\right)^{1/\alpha} \right\}.$$

All subsequent local contraction and linear-convergence statements hold with $r_0(\tau)$ replaced by $r_0^{\text{gap}}(\tau)$ and $\beta_{\text{loc}}(\tau)$ replaced by $\beta^{\text{gap}}_{\text{loc}}(\tau)$, *informally extending the same arguments*, provided the contraction ball around $Q^\star_{\mathcal{F}}(\tau)$ remains inside $\mathbb{B}_{\text{gap}}$.

A possible route toward global convergence. By combining the $\tau > 1$ and $\tau < 1$ regimes, one obtains a continuous temperature– annealing path. One can take τ sufficiently large to ensure initialization within a contraction region and then decrease τ gradually. As $\tau \downarrow 0$, the contraction region stabilizes to $\mathbb{B}_{\text{gap}}$. Heuristically, if the fitted soft Q-iteration iterates remain within the appropriate local basin for all intermediate temperatures, the solution should converge even at temperatures arbitrarily close to the hard-optimal limit.

Formalizing this continuation argument presents several challenges. Our current bounds are expressed in the stationary norm induced by the τ-dependent soft-optimal policy, which varies with temperature. A full temperature–annealing analysis would require a common, temperature-independent norm; the stationary norm associated with the hard-max optimal policy is a natural candidate for controlling the entire path.

C. Gradient-Descent Interpretation

A useful perspective on the projected soft Bellman update is obtained by writing it in residual form. Define

$$F_{\mathcal{F}}(Q) := Q - \mathcal{T}_{\mathcal{F}}(Q),$$

so the population iteration becomes

$$Q^{(k+1)} = Q^{(k)} - F_{\mathcal{F}}(Q^{(k)}).$$

This is a unit-step projected gradient method, and its behavior near $Q^\star$ is governed by the local geometry of $F_{\mathcal{F}}$.

Suppose for simplicity that $Q^\star$ lies in the interior of $\mathcal{F}$, so that the projection is smooth in a neighborhood of $Q^\star$. At the soft-optimal fixed point $Q^\star$, the Jacobian of $F_{\mathcal{F}}$ is

$$DF_{\mathcal{F}}(Q^\star) = I - \gamma\, \Pi_{\mathcal{F}}\, T^{\text{eval}}_{\pi^\star}.$$

Because $T^{\text{eval}}_{\pi^\star}$ is a γ-contraction in the stationary norm and $\Pi_{\mathcal{F}}$ is nonexpansive in this geometry, the operator $\gamma\, \Pi_{\mathcal{F}} T^{\text{eval}}_{\pi^\star}$ has norm at most γ. Consequently, $DF_{\mathcal{F}}(Q^\star)$ is positive definite on the action-differential subspace, implying that $F_{\mathcal{F}}$ is locally *strongly monotone* and smooth in a neighborhood of $Q^\star$ (Bauschke & Combettes, 2011). In this region, the projected soft Bellman update behaves like gradient descent on a locally strongly convex, smooth objective.

D. Proofs for Operator Smoothness Results

D.1. First derivative of Bellman Operator

Proof of Lemma 1. Fix (s, a) and a perturbation H. Since the reward term does not depend on Q,

$$D\mathcal{T}(Q)[H](s,a) = \gamma\, \mathbb{E}_{S'\sim P(\cdot|s,a)}\big[D\phi_{S'}(Q)[H]\big],$$

where $\phi_{S'}(Q) := \tau \log \sum_b \exp(Q(S', b)/\tau)$. Let $g(z) = \tau \log \sum_b e^{z_b/\tau}$ with $z_b(Q) = Q(S', b)$. The gradient of g is the softmax, so $\partial g/\partial z_b(z(Q)) = \pi_Q(b \mid S')$. By the chain rule,

$$D\phi_{S'}(Q)[H] = \sum_b \pi_Q(b \mid S')\, H(S', b).$$

Substituting back gives

$$\begin{aligned} D\mathcal{T}(Q)[H](s,a) &= \gamma\, \mathbb{E}_{S'\sim P(\cdot|s,a)}\Big[\sum_b \pi_Q(b \mid S')\, H(S', b)\Big] \\ &= \gamma\, P^{\text{eval}}_{\pi_Q} H(s,a). \end{aligned}$$

which proves the claim. □

D.2. Second derivative and local curvature

Computation of second derivative. The next lemma shows that the second derivative of $\mathcal{T}$ has a positive–semidefinite covariance form.

Lemma 6 (Second derivative)**.** *For any Q and directions H_1, H_2,*

$$D^2\mathcal{T}(Q)[H_1, H_2](s,a) = \gamma\, \mathbb{E}_{S'\sim P(\cdot|s,a)}\left[\tfrac{1}{\tau}\operatorname{Cov}_{\pi_Q}\big(H_1(S',\cdot),\, H_2(S',\cdot)\big)\right].$$

Proof of Lemma 6. Fix (s, a) and directions H_1, H_2. As in Lemma 1,

$$(\mathcal{T}Q)(s,a) = r_0(s,a) + \gamma\, \mathbb{E}_{S'\sim P(\cdot|s,a)}[\phi_{S'}(Q)],$$
$$\phi_{S'}(Q) := \tau \log \sum_b \exp\big(Q(S', b)/\tau\big).$$

Thus,

$$\begin{aligned} &D^2\mathcal{T}(Q)[H_1, H_2](s,a) \\ &= \gamma\, \mathbb{E}_{S'\sim P(\cdot|s,a)}\big[D^2\phi_{S'}(Q)[H_1, H_2]\big]. \end{aligned}$$

For fixed S', write $z_b(Q) = Q(S', b)$ and $g(z) = \tau \log \sum_b e^{z_b/\tau}$, so $\phi_{S'}(Q) = g(z(Q))$. Let $\pi_Q(\cdot \mid S')$ be the softmax at $z(Q)$. Then

$$\begin{aligned} \frac{\partial g}{\partial z_b}\big(z(Q)\big) &= \pi_Q(b \mid S'), \\ \frac{\partial^2 g}{\partial z_b\, \partial z_c}\big(z(Q)\big) &= \frac{1}{\tau}\Big[\pi_Q(b \mid S')\, \mathbf{1}\{b = c\} \\ &\qquad - \pi_Q(b \mid S')\, \pi_Q(c \mid S')\Big]. \end{aligned}$$

Hence

$$\begin{aligned} &D^2\phi_{S'}(Q)[H_1, H_2] \\ &= \sum_{b,c} \frac{\partial^2 g}{\partial z_b\, \partial z_c}(z(Q))\, H_1(S', b)\, H_2(S', c) \\ &= \frac{1}{\tau}\Big(\mathbb{E}_{\pi_Q}[H_1(S',\cdot)\, H_2(S',\cdot)] \\ &\qquad - \mathbb{E}_{\pi_Q}[H_1(S',\cdot)]\, \mathbb{E}_{\pi_Q}[H_2(S',\cdot)]\Big) \\ &= \frac{1}{\tau}\operatorname{Cov}_{\pi_Q}\big(H_1(S',\cdot),\, H_2(S',\cdot)\big). \end{aligned}$$

Taking expectation over S' yields the claimed expression.

For positive semidefiniteness, set $H_1 = H_2 = H$:

$$\begin{aligned} D^2\mathcal{T}(Q)[H, H](s,a) &= \frac{\gamma}{\tau}\, \mathbb{E}_{S'\sim P(\cdot|s,a)}\Big[\operatorname{Var}_{\pi_Q}\big(H(S',\cdot)\big)\Big] \\ &\geq 0. \end{aligned}$$

In particular, $D^2\mathcal{T}(Q)$ is positive semidefinite for every Q. Moreover, for any fixed S' the covariance form is strictly positive whenever $H(S', \cdot)$ is nonconstant under $\pi_Q(\cdot \mid S')$, so $D^2\mathcal{T}(Q)$ is strictly positive definite along nontrivial action-differential directions. □

Uniform boundedness of second derivative and curvature bound. In the following lemma, we define

$$\mathcal{F}_{\text{diff}} := \{\, Q_1 - Q_2 :\ Q_1, Q_2 \in \mathcal{F} \cup \mathcal{T}(\mathcal{F}) \cup \{Q^\star\}\,\}.$$

Lemma 7 (Uniform boundedness of the second derivative)**.** *Assume Conditions C1–C3 hold. Then there exists a finite constant*

$$\beta_{\text{loc}} := \frac{\gamma}{2\tau}\, C_\infty \sqrt{\frac{|\mathcal{A}|}{\pi_{\min}}}$$

such that, for all Q and all directions $H_1 \in \mathcal{F}_{\text{diff}}$ and $H_2 \in L^2(\mu^*)$,

$$\big\|D^2\mathcal{T}(Q)[H_1, H_2]\big\|_{2,\mu^\star} \;\leq\; \beta_{\text{loc}}\, \|H_1\|^{\alpha}_{2,\mu^\star}\, \|H_2\|_{2,\mu^\star}, \tag{6}$$

where α is the exponent from C3.

Proof. By Lemma 6, $D^2\mathcal{T}(Q)[H_1, H_2]$ can be expressed in terms of the covariance $\operatorname{Cov}_{\pi_Q}\big(H_1(S',\cdot),\, H_2(S',\cdot)\big)$.

For fixed S', write $h_i := H_i(S', \cdot) \in \mathbb{R}^{|\mathcal{A}|}$ and $p_Q := \pi_Q(\cdot \mid S')$. Then

$$\operatorname{Cov}_{\pi_Q}\big(H_1(S',\cdot), H_2(S',\cdot)\big) = h_1^\top\big(\operatorname{diag}(p_Q) - p_Q p_Q^\top\big) h_2,$$

and the covariance matrix $\operatorname{diag}(p_Q) - p_Q p_Q^\top$ has spectral norm at most $1/2$, so

$$\big|\operatorname{Cov}_{\pi_Q}\big(H_1(S',\cdot), H_2(S',\cdot)\big)\big| \;\leq\; \tfrac{1}{2}\, \|h_1\|_2\, \|h_2\|_2.$$

Thus, for each (s, a),

$$\begin{aligned}&\left|D^2\mathcal{T}(Q)[H_1, H_2](s, a)\right| \\ &\leq \frac{\gamma}{2\tau}\,\mathbb{E}_{S'|s,a}\left[\,\|H_1(S', \cdot)\|_2\,\|H_2(S', \cdot)\|_2\,\right].\end{aligned}$$

Let $(S, A) \sim \mu^\star$ and $S' \sim P(\cdot \mid S, A)$, and define $Z(S') := \|H_1(S', \cdot)\|_2\,\|H_2(S', \cdot)\|_2$. By Jensen's inequality,

$$\begin{aligned}\left\|D^2\mathcal{T}(Q)[H_1, H_2]\right\|_{2,\mu^\star}^2 &= \mathbb{E}_{\mu^\star}\left[D^2\mathcal{T}(Q)[H_1, H_2](S, A)^2\right] \\ &\leq \left(\frac{\gamma}{2\tau}\right)^2 \mathbb{E}_{\mu^\star}\left[\mathbb{E}_{S'|S,A} Z(S')^2\right].\end{aligned}$$

Since $\mu^\star$ is stationary for $\pi^\star$, the marginal distribution of S' under $(S, A) \sim \mu^\star$, $S' \sim P(\cdot \mid S, A)$ coincides with the state marginal of $\mu^\star$. Hence

$$\mathbb{E}_{\mu^\star}\left[\mathbb{E}_{S'|S,A} Z(S')^2\right] = \mathbb{E}_{S'}\left[\|H_1(S', \cdot)\|_2^2\,\|H_2(S', \cdot)\|_2^2\right].$$

By the assumption $\pi^\star(a \mid s) \geq \pi_{\min}$ whenever $\mu^\star(s, a) > 0$, for any state s and $h \in \mathbb{R}^{|\mathcal{A}|}$,

$$\begin{aligned}\|h\|_2^2 &= \sum_a h(a)^2 \\ &= \sum_a \frac{h(a)^2}{\pi^\star(a \mid s)}\,\pi^\star(a \mid s) \\ &\leq \frac{1}{\pi_{\min}} \sum_a \pi^\star(a \mid s)\,h(a)^2.\end{aligned}$$

Applying this with $h = H_i(S', \cdot)$ gives

$$\|H_i(S', \cdot)\|_2^2 \;\leq\; \frac{1}{\pi_{\min}} \sum_a \pi^\star(a \mid S')\,H_i(S', a)^2.$$

Conditions C1–C3, and using $\|H_1(S', \cdot)\|_2^2 \leq \sup_s \|H_1(s, \cdot)\|_2^2 \leq |\mathcal{A}|\,\|H_1\|_{\infty,\mu^\star}^2$, we obtain

$$\begin{aligned}&\mathbb{E}_{S'}\left[\|H_1(S', \cdot)\|_2^2\,\|H_2(S', \cdot)\|_2^2\right] \\ &\quad\leq \sup_s \|H_1(s, \cdot)\|_2^2\,\mathbb{E}_{S'}\left[\|H_2(S', \cdot)\|_2^2\right] \\ &\quad\leq |\mathcal{A}|\,\|H_1\|_{\infty,\mu^\star}^2\,\mathbb{E}_{S'}\left[\|H_2(S', \cdot)\|_2^2\right] \\ &\quad\leq \frac{|\mathcal{A}|}{\pi_{\min}}\,\|H_1\|_{\infty,\mu^\star}^2\,\|H_2\|_{2,\mu^\star}^2 \\ &\quad\leq \frac{|\mathcal{A}|}{\pi_{\min}}\,C_\infty^2\,\|H_1\|_{2,\mu^\star}^{2\alpha}\,\|H_2\|_{2,\mu^\star}^2.\end{aligned}$$

In the final inequality, we crucially used that $H_1 \in \mathcal{F}_{\text{diff}}$ together with the sup–L^2 coupling in C3. Combining with the previous display gives

$$\begin{aligned}&\left\|D^2\mathcal{T}(Q)[H_1, H_2]\right\|_{2,\mu^\star}^2 \\ &\quad\leq \frac{|\mathcal{A}|}{\pi_{\min}}\,C_\infty^2\left(\frac{\gamma}{2\tau}\right)^2 \|H_1\|_{2,\mu^\star}^{2\alpha}\,\|H_2\|_{2,\mu^\star}^2 \\ &\quad= \frac{|\mathcal{A}|\,C_\infty^2}{\pi_{\min}}\left(\frac{\gamma}{2\tau}\right)^2 \|H_1\|_{2,\mu^\star}^{2\alpha}\,\|H_2\|_{2,\mu^\star}^2.\end{aligned}$$

so that

$$\left\|D^2\mathcal{T}(Q)[H_1, H_2]\right\|_{2,\mu^\star} \;\leq\; \beta_{\text{loc}}\,\|H_1\|_{2,\mu^\star}^{\alpha}\,\|H_2\|_{2,\mu^\star},$$

where $\beta_{\text{loc}} := \frac{\gamma}{2\tau}\,C_\infty\sqrt{\frac{|\mathcal{A}|}{\pi_{\min}}}$. The bound does not depend on Q. Setting $Q = Q^\star$ gives the lemma statement.

We note that Condition C3 can be relaxed to the weaker L^4–L^2 coupling $\|H\|_{4,\mu^\star}^2 \leq C_4\,\|H\|_{2,\mu^\star}^{1+\alpha}$, which accommodates function classes that are not uniformly bounded in $\|\cdot\|_\infty$, provided that this bound holds uniformly for all directions H_1 and H_2 used in the local analysis. □

Proof of Lemma 2. Fix Q and define $Q_t := Q^\star + t(Q - Q^\star)$ for $t \in [0, 1]$. By the second-order Taylor expansion in Banach spaces with integral remainder,

$$\mathcal{T}(Q) - \mathcal{T}(Q^\star) = D\mathcal{T}(Q^\star)[Q - Q^\star] + \int_0^1 (1-t)\,D^2\mathcal{T}(Q_t)[Q - Q^\star,\, Q - Q^\star] \tag{7}$$

By Lemma 1,

$$D\mathcal{T}(Q^\star)[Q - Q^\star] = \gamma\,P_{\pi^\star}^{\text{eval}}(Q - Q^\star).$$

Since $Q^\star$ is a fixed point, $\mathcal{T}(Q^\star) = Q^\star$, and

$$\mathcal{T}(Q^\star) + D\mathcal{T}(Q^\star)[Q - Q^\star] = Q^\star + \gamma P_{\pi^\star}^{\text{eval}}(Q - Q^\star) = \mathcal{T}_{Q^\star}^{\text{eval}}(Q),$$

substituting into (7) yields

$$\mathcal{T}(Q) - \mathcal{T}_{Q^\star}^{\text{eval}}(Q) = \int_0^1 (1-t)\,D^2\mathcal{T}(Q_t)[Q - Q^\star,\, Q - Q^\star]\,dt.$$

Applying Lemma 7 with $H_1 = H_2 = Q - Q^\star$, for each $t \in [0, 1]$,

$$\left\|D^2\mathcal{T}(Q_t)[Q - Q^\star,\, Q - Q^\star]\right\|_{2,\mu^\star} \leq \beta_{\text{loc}}\,\|Q - Q^\star\|_{2,\mu^\star}^{1+\alpha}.$$

Therefore,

$$\begin{aligned}\left\|\mathcal{T}(Q) - \mathcal{T}_{Q^\star}^{\text{eval}}(Q)\right\|_{2,\mu^\star} &\leq \int_0^1 (1-t)\left\|D^2\mathcal{T}(Q_t)[Q - Q^\star,\, Q - Q^\star]\right\|_{2,\mu^\star} \\ &\leq \left(\int_0^1 (1-t)\,dt\right)\beta_{\text{loc}}\,\|Q - Q^\star\|_{2,\mu^\star}^{1+\alpha} \\ &= \frac{\beta_{\text{loc}}}{2}\,\|Q - Q^\star\|_{2,\mu^\star}^{1+\alpha}.\end{aligned}$$

□

D.3. Better temperature dependence of curvature under margin conditions.

The next lemma refines the previous curvature bound under additional structural assumptions. Under a uniform action gap and within a suitably small $L^2(\mu^\star)$-ball around $Q^\star$, the second derivative of $\mathcal{T}_\tau$ admits an improved bound of order $\frac{1}{\tau}e^{-\Delta/(2\tau)}$. This refinement is strictly local around $Q^\star$ and strengthens the curvature bound of Lemma 7, which is used throughout the main text.

Lemma 8 (Local second-derivative bound under an action gap)**.** *Assume Conditions C1–C3 and the structural assumptions S1–S2. Then there exists a finite constant*

$$\beta_{\mathrm{loc}}^{\mathrm{gap}}(\tau) := \frac{\gamma}{\tau} C_\infty \sqrt{\frac{|\mathcal{A}|}{\pi_{\min}}} C_{\mathrm{gap}} e^{-\Delta/(2\tau)}$$

for some $C_{\mathrm{gap}} < \infty$ such that, for all $Q \in \mathbb{B}_{\mathrm{gap}}$ and all directions $H_1 \in \mathcal{F}_{\mathrm{diff}}$, $H_2 \in L^2(\mu^\star)$,

$$\left\|D^2\mathcal{T}(Q)[H_1, H_2]\right\|_{2,\mu^\star} \;\le\; \beta_{\mathrm{loc}}^{\mathrm{gap}}(\tau)\, \|H_1\|_{2,\mu^\star}^{\alpha}\, \|H_2\|_{2,\mu^\star}. \tag{8}$$

In particular, for fixed $\Delta > 0$ the local curvature decays as $\beta_{\mathrm{loc}}^{\mathrm{gap}}(\tau) = O\big(\frac{1}{\tau} e^{-\Delta/(2\tau)}\big)$ as $\tau \downarrow 0$.

Proof. The proof follows the same structure as Lemma 7, with an improved bound on the covariance term under the action-gap condition. By Lemma 6, $D^2\mathcal{T}(Q)[H_1, H_2]$ can be written in terms of $\mathrm{Cov}_{\pi_Q}\big(H_1(S', \cdot), H_2(S', \cdot)\big)$. Fix S' and write $h_i := H_i(S', \cdot) \in \mathbb{R}^{|\mathcal{A}|}$ and $p_Q := \pi_Q(\cdot \mid S')$. The gap assumption at $Q^\star_{\mathrm{hard}}$, together with Condition C3 and the choice of r_{gap}, implies that for all $Q \in \mathbb{B}_{\mathrm{gap}}$ the same action $A^\star(S')$ remains uniquely optimal and

$$Q(S', A^\star(S')) \;\ge\; Q(S', a) + \Delta \quad \text{for all } a \neq A^\star(S').$$

By the definition of the softmax policy,

$$p_Q(a) = \frac{\exp\{Q(S', a)/\tau\}}{\sum_{b \in \mathcal{A}} \exp\{Q(S', b)/\tau\}}, \qquad a \in \mathcal{A}.$$

For any $a \neq A^\star(S')$ we have

$$\frac{p_Q(a)}{p_Q(A^\star(S'))} = \exp\left(\frac{Q(S', a) - Q(S', A^\star(S'))}{\tau}\right) \;\le\; e^{-\Delta/\tau},$$

so $p_Q(a) \le e^{-\Delta/\tau}$ and

$$\sum_{a \neq A^\star(S')} p_Q(a) \;\le\; (|\mathcal{A}| - 1) e^{-\Delta/\tau} =: \varepsilon.$$

Thus $p_Q(A^\star(S')) \ge 1 - \varepsilon$, and we may take $C'_{\mathrm{gap}} := |\mathcal{A}| - 1$ so that $p_Q(A^\star(S')) \ge 1 - C'_{\mathrm{gap}} e^{-\Delta/\tau}$.

Next, consider the covariance matrix $\Sigma_Q := \mathrm{diag}(p_Q) - p_Q p_Q^\top$. For any $h \in \mathbb{R}^{|\mathcal{A}|}$,

$$h^\top \Sigma_Q h = \mathrm{Var}_{\pi_Q}\big(h(A)\big).$$

Let $a^\star := A^\star(S')$ and write $\varepsilon := \sum_{a \neq a^\star} p_Q(a)$ as above, and set $m := \mathbb{E}_{\pi_Q}[h(A)]$. Then

$$\begin{aligned}
\mathrm{Var}_{\pi_Q}(h) &= \sum_a p_Q(a)\big(h(a) - m\big)^2 \\
&\le 2 \sum_a p_Q(a)\big(h(a) - h(a^\star)\big)^2 + 2\big(h(a^\star) - m\big)^2 \\
&\le 4 \sum_{a \neq a^\star} p_Q(a)\big(h(a) - h(a^\star)\big)^2 \\
&\le C_{\mathrm{gap}}\, \varepsilon\, \|h\|_2^2,
\end{aligned}$$

where we used $(x - y)^2 \le 2(x^2 + y^2)$ and absorbed numerical constants into C_{gap}. Since $\varepsilon \le C'_{\mathrm{gap}} e^{-\Delta/\tau}$, we obtain

$$\mathrm{Var}_{\pi_Q}(h) \;\le\; C_{\mathrm{gap}} e^{-\Delta/\tau}\, \|h\|_2^2.$$

Using $e^{-\Delta/\tau} \le e^{-\Delta/(2\tau)}$ for $\tau > 0$, we can absorb the resulting change into the constant and write

$$\mathrm{Var}_{\pi_Q}(h) \;\le\; C_{\mathrm{gap}} e^{-\Delta/(2\tau)}\, \|h\|_2^2.$$

Applying Cauchy–Schwarz to the covariance,

$$\begin{aligned}
\big|\mathrm{Cov}_{\pi_Q}(h_1, h_2)\big| &\le \sqrt{\mathrm{Var}_{\pi_Q}(h_1)\ \mathrm{Var}_{\pi_Q}(h_2)} \\
&\le\; C_{\mathrm{gap}} e^{-\Delta/(2\tau)}\, \|h_1\|_2\, \|h_2\|_2.
\end{aligned}$$

Substituting this bound into the argument of Lemma 7, and repeating the same steps (Jensen's inequality, stationarity of $\mu^\star$, and the $\pi_{\min}$-lower bound), we obtain

$$\begin{aligned}
&\left\|D^2\mathcal{T}(Q)[H_1, H_2]\right\|_{2,\mu^\star}^2 \\
&\quad\le\; \frac{|\mathcal{A}|}{\pi_{\min}} C_\infty^2 \left(\frac{\gamma}{\tau} C_{\mathrm{gap}} e^{-\Delta/(2\tau)}\right)^2 \|H_1\|_{2,\mu^\star}^{2\alpha}\, \|H_2\|_{2,\mu^\star}^2,
\end{aligned}$$

for all $Q \in \mathbb{B}_{\mathrm{gap}}$, which yields (8) with the stated $\beta_{\mathrm{loc}}^{\mathrm{gap}}(\tau)$. The bound is uniform over $Q \in \mathbb{B}_{\mathrm{gap}}$ and depends on τ only through the prefactor $\frac{1}{\tau} e^{-\Delta/(2\tau)}$. □

E. Proof of local contraction theorem

Proof of Theorem 1. First, since $r > \mathrm{dist}(\mathcal{F}, Q^\star)$, we have that $\mathbb{B}_{\mathcal{F}}(r, Q^\star)$ is the intersection of a closed ball (which is convex) and a convex set. It is therefore also convex. It is also nonempty because $r > \mathrm{dist}(\mathcal{F}, Q^\star)$ and $\Pi_{\mathcal{F}} Q^\star \in \mathcal{F}$ with $\|\Pi_{\mathcal{F}} Q^\star - Q^\star\|_{2,\mu^\star} = \mathrm{dist}(\mathcal{F}, Q^\star) < r$. Fix $Q_1, Q_2 \in \mathbb{B}_{\mathcal{F}}(r, Q^\star)$ and set $H := Q_1 - Q_2$. For $t \in [0, 1]$, define the path $Q_t := Q_2 + tH \in \mathcal{F}$. By the fundamental theorem of calculus in Banach spaces,

$$\mathcal{T}(Q_1) - \mathcal{T}(Q_2) = \int_0^1 D\mathcal{T}(Q_t)[H]\, dt.$$

Taking $L^2(\mu^\star)$ norms and using the triangle inequality,

$$\|\mathcal{T}(Q_1) - \mathcal{T}(Q_2)\|_{2,\mu^\star} \;\le\; \int_0^1 \|D\mathcal{T}(Q_t)[H]\|_{2,\mu^\star}\, dt. \tag{9}$$

We now bound $\|D\mathcal{T}(Q_t)[H]\|_{2,\mu^\star}$ uniformly over $t \in [0, 1]$ and $Q_t \in B_{\mathcal{F}}(r, Q^\star)$. Add and subtract the derivative at the optimum:

$$D\mathcal{T}(Q_t)[H] = D\mathcal{T}(Q^\star)[H] + \big(D\mathcal{T}(Q_t) - D\mathcal{T}(Q^\star)\big)[H].$$

Using Lemma 1 and Lemma 7, we control each term separately.

First, at $Q^\star$,

$$\|D\mathcal{T}(Q^\star)[H]\|_{2,\mu^\star} = \gamma \|P^{\text{eval}}_{\pi^\star} H\|_{2,\mu^\star} \;\le\; \gamma \|H\|_{2,\mu^\star}, \tag{10}$$

since $P^{\text{eval}}_{\pi^\star}$ is nonexpansive in $L^2(\mu^\star)$.

Second, by the mean–value representation of the derivative,

$$\begin{aligned}&\big(D\mathcal{T}(Q_t) - D\mathcal{T}(Q^\star)\big)[H] \\ &= \int_0^1 D^2\mathcal{T}(Q^\star + u\,(Q_t - Q^\star))\,\big[\,Q_t - Q^\star,\; H\,\big]\,du.\end{aligned}$$

Applying Lemma 7 with $H_1 = Q_t - Q^\star \in \mathcal{F}_{\text{diff}}$ and $H_2 = H \in L^2(\mu^\star)$ gives

$$\begin{aligned}&\Big\|\big(D\mathcal{T}(Q_t) - D\mathcal{T}(Q^\star)\big)[H]\Big\|_{2,\mu^\star} \\ &\le \int_0^1 \beta_{\text{loc}}\,\|Q_t - Q^\star\|^\alpha_{2,\mu^\star}\,\|H\|_{2,\mu^\star}\,du.\end{aligned}$$

where we used that $Q_t - Q^\star \in \mathcal{F}_{\text{diff}}$ because $Q_t \in \mathcal{F}$. Since Q_t lies on the segment between Q_1 and Q_2, both in $B_{\mathcal{F}}(r, Q^\star)$,

$$\|Q_t - Q^\star\|_{2,\mu^\star} \;\le\; \max\big\{\|Q_1 - Q^\star\|_{2,\mu^\star}, \|Q_2 - Q^\star\|_{2,\mu^\star}\big\} \;\le\; r$$

Thus,

$$\Big\|\big(D\mathcal{T}(Q_t) - D\mathcal{T}(Q^\star)\big)[H]\Big\|_{2,\mu^\star} \;\le\; \beta_{\text{loc}}\, r^\alpha\, \|H\|_{2,\mu^\star}. \tag{11}$$

Combining the bounds in (10) and (11), we obtain, for all $t \in [0,1]$,

$$\|D\mathcal{T}(Q_t)[H]\|_{2,\mu^\star} \;\le\; \gamma\,\|H\|_{2,\mu^\star} + \beta_{\text{loc}}\, r^\alpha\, \|H\|_{2,\mu^\star} = \rho(r)\,\|H\|_{2,\mu^\star}$$

where $\rho(r) := \gamma + \beta_{\text{loc}} r^\alpha$. Substituting this bound into (9) yields

$$\|\mathcal{T}(Q_1) - \mathcal{T}(Q_2)\|_{2,\mu^\star} \;\le\; \int_0^1 \rho(r)\,\|H\|_{2,\mu^\star}\,dt = \rho(r)\,\|Q_1 - Q_2\|_{2,\mu^\star}.$$

Finally, by construction

$$r < r_0 = \left(\frac{1-\gamma}{\beta_{\text{loc}}}\right)^{1/\alpha} \quad\Longrightarrow\quad \rho(r) = \gamma + \beta_{\text{loc}} r^\alpha < 1,$$

so $\mathcal{T}$ is a strict contraction on $B_{\mathcal{F}}(r, Q^\star)$ in the $L^2(\mu^\star)$ norm. □

E.1. Proof of locally unique existence of fixed point

Proof of Lemma 3. Fix $r \in (0, r_0)$ satisfying (3), and let $Q \in \mathbb{B}_{\mathcal{F}}(r, Q^\star)$. By Theorem 1, the operator $\mathcal{T}_{\mathcal{F}}$ is a $\rho(r)$-contraction on $\mathbb{B}_{\mathcal{F}}(r, Q^\star)$, so in particular

$$\|\mathcal{T}_{\mathcal{F}}(Q) - \mathcal{T}_{\mathcal{F}}(Q^\star)\|_{2,\mu^\star} \;\le\; \rho(r)\,\|Q - Q^\star\|_{2,\mu^\star} \;\le\; \rho(r)\,r.$$

Moreover, since $\mathcal{T}(Q^\star) = Q^\star$, we have $\mathcal{T}_{\mathcal{F}}(Q^\star) = \Pi_{\mathcal{F}} Q^\star$ and hence

$$\|\mathcal{T}_{\mathcal{F}}(Q^\star) - Q^\star\|_{2,\mu^\star} = \|\Pi_{\mathcal{F}} Q^\star - Q^\star\|_{2,\mu^\star} = \text{dist}(\mathcal{F}, Q^\star).$$

By the triangle inequality,

$$\begin{aligned}&\|\mathcal{T}_{\mathcal{F}}(Q) - Q^\star\|_{2,\mu^\star} \\ &\le \|\mathcal{T}_{\mathcal{F}}(Q) - \mathcal{T}_{\mathcal{F}}(Q^\star)\|_{2,\mu^\star} + \|\mathcal{T}_{\mathcal{F}}(Q^\star) - Q^\star\|_{2,\mu^\star} \\ &\le \rho(r)\,r + \text{dist}(\mathcal{F}, Q^\star) \\ &\le \rho(r)\,r + (1 - \rho(r))\,r \;=\; r,\end{aligned}$$

where the last inequality uses (3). Thus $\mathcal{T}_{\mathcal{F}}(Q) \in \mathbb{B}_{\mathcal{F}}(r, Q^\star)$, so $\mathcal{T}_{\mathcal{F}}$ maps $\mathbb{B}_{\mathcal{F}}(r, Q^\star)$ into itself.

By Theorem 1, $\mathcal{T}_{\mathcal{F}}$ is a strict contraction on $\mathbb{B}_{\mathcal{F}}(r, Q^\star)$ with respect to $\|\cdot\|_{2,\mu^\star}$. Since $\mathbb{B}_{\mathcal{F}}(r, Q^\star)$ is a closed, convex subset of $L^2(\mu^\star)$ and is nonempty for $r > \text{dist}(\mathcal{F}, Q^\star)$, Banach's fixed-point theorem implies that $\mathcal{T}_{\mathcal{F}}$ admits a unique fixed point $Q^\star_{\mathcal{F}}$ in $\mathbb{B}_{\mathcal{F}}(r, Q^\star)$. □

E.2. Proof of local linear convergence

Lemma 9. *Suppose Condition C4 holds. Then there exists* $r \in (0, r_0)$ *such that*

$$\text{dist}(\mathcal{F}, Q^\star) \;\le\; (1 - \rho(r))\,r,$$

so the assumptions of Lemma 3 are satisfied.

Proof. Recall $\rho(r) := \gamma + \beta_{\text{loc}} r^\alpha$ and define

$$r_* := \left(\frac{1-\gamma}{(\alpha+1)\,\beta_{\text{loc}}}\right)^{1/\alpha}.$$

Then $r_* \in (0, r_0)$ since $r_*^\alpha = (1-\gamma)/\big((\alpha+1)\beta_{\text{loc}}\big) < (1-\gamma)/\beta_{\text{loc}} = r_0^\alpha$. Moreover,

$$\beta_{\text{loc}} r_*^\alpha = \frac{1-\gamma}{\alpha+1}, \qquad \rho(r_*) = \gamma + \frac{1-\gamma}{\alpha+1},$$

$$1 - \rho(r_*) = 1 - \gamma - \frac{1-\gamma}{\alpha+1} = \frac{\alpha(1-\gamma)}{\alpha+1},$$

and therefore

$$(1 - \rho(r_*))\,r_* = \frac{\alpha}{\alpha+1}\,(1-\gamma)\left(\frac{1-\gamma}{(\alpha+1)\,\beta_{\text{loc}}}\right)^{1/\alpha}.$$

By Condition C4, $\varepsilon_{\mathcal{F}} \le (1 - \rho(r_*))\,r_*$. Since $Q^\star_{\mathcal{F}} \in \mathcal{F}$ and $\varepsilon_{\mathcal{F}} = \|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu^\star}$, we have

$$\text{dist}(\mathcal{F}, Q^\star) \;\le\; \varepsilon_{\mathcal{F}} \;\le\; (1 - \rho(r_*))\,r_*.$$

Thus the invariance condition of Lemma 3 holds with $r = r_* \in (0, r_0)$. □

Proof of Theorem 2. By Condition C4 and Lemma 9, the projected fixed point $Q^\star_{\mathcal{F}}$ exists.

Step 1: Contraction on a neighborhood of $Q^\star_{\mathcal{F}}$. For any $Q \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$, the triangle inequality gives

$$\|Q - Q^\star\|_{2,\mu^\star} \;\le\; \|Q - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} + \|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu^\star} \;\le\; r + \varepsilon_{\mathcal{F}}.$$

Thus

$$\mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}}) \;\subseteq\; \mathbb{B}_{\mathcal{F}}(r + \varepsilon_{\mathcal{F}}, Q^\star).$$

Since $\varepsilon_{\mathcal{F}} \ge \mathrm{dist}(\mathcal{F}, Q^\star)$ and $r + \varepsilon_{\mathcal{F}} < r_0$, the local contraction property of Theorem 1 applies with radius $R := r + \varepsilon_{\mathcal{F}}$ and yields: for all $Q_1, Q_2 \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$

$$\|\mathcal{T}_{\mathcal{F}}(Q_1) - \mathcal{T}_{\mathcal{F}}(Q_2)\|_{2,\mu^\star} \;\le\; \rho(R)\, \|Q_1 - Q_2\|_{2,\mu^\star},$$

where

$$\rho(R) = \gamma + \beta_{\mathrm{loc}} R^\alpha = \rho_{\mathrm{eff}}(r).$$

Step 2: Invariance and linear convergence. Let $Q^{(0)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$ and define $Q^{(k+1)} = \mathcal{T}_{\mathcal{F}}(Q^{(k)})$. Since $Q^\star_{\mathcal{F}} = \mathcal{T}_{\mathcal{F}}(Q^\star_{\mathcal{F}})$, by Theorem 1, we have for any $Q \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$,

$$\begin{aligned}\|\mathcal{T}_{\mathcal{F}}(Q) - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} &= \|\mathcal{T}_{\mathcal{F}}(Q) - \mathcal{T}_{\mathcal{F}}(Q^\star_{\mathcal{F}})\|_{2,\mu^\star} \\ &\le\; \rho_{\mathrm{eff}}(r)\, \|Q - Q^\star_{\mathcal{F}}\|_{2,\mu^\star}.\end{aligned}$$

Since $\rho_{\mathrm{eff}}(r) < 1$, it follows that $\mathcal{T}_{\mathcal{F}}(Q) \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$, so the ball is invariant.

By induction, $Q^{(k)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$ for all $k \ge 0$, and

$$\|Q^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \;\le\; \rho_{\mathrm{eff}}(r)^k\, \|Q^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star},$$

which proves the claim.

□

Proof of Corollary 1. Under the assumption $Q^\star_{\mathcal{F}} = Q^\star$ we have $\varepsilon_{\mathcal{F}} = 0$, so Theorem 2 yields for any $Q^{(0)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star)$ and all $k \ge 0$,

$$e_k = \|Q^{(k)} - Q^\star\|_{2,\mu^\star} \;\le\; \rho^k\, e_0, \quad \rho := \gamma + \beta_{\mathrm{loc}} r^\alpha.$$

By the local contraction bound (applied at $Q^{(k)}$),

$$e_{k+1} \;\le\; \big(\gamma + \beta_{\mathrm{loc}}\, e_k^\alpha\big)\, e_k,$$

so the per-iteration modulus

$$\rho_k := \frac{e_{k+1}}{e_k}$$

satisfies

$$\rho_k - \gamma \;\le\; \beta_{\mathrm{loc}}\, e_k^\alpha.$$

Using the bound from Theorem 2,

$$e_k^\alpha \;\le\; (\rho^k e_0)^\alpha = e_0^\alpha\, \rho^{\alpha k}.$$

Combining the last two displays gives

$$\rho_k - \gamma \;\le\; \beta_{\mathrm{loc}}\, e_0^\alpha\, \rho^{\alpha k}, \qquad k \ge 0,$$

which is the claimed finite-time tightening of the contraction rate. □

F. Proofs for inexact Picard iteration

Proof of Lemma 4. Let $\mathcal{T} := \mathcal{T}_{\mathcal{F}}$ and

$$e_k := \|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star}, \qquad k \ge 0.$$

By Theorem 2, for any $0 < r < r_{\max}$ the operator $\mathcal{T}$ is a contraction on the ball

$$\mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}}) := \{Q \in \mathcal{F} : \|Q - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \le r\}$$

with modulus $\rho_\star = \rho_{\mathrm{eff}}(r)$, that is,

$$\|\mathcal{T}(Q) - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \le \rho_\star \|Q - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \quad \text{for all } Q \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}}).$$

Step 1: Staying inside the basin. We first show by induction that $e_k \le r$ for all $k \ge 0$, so that $\widehat{Q}^{(k)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$ and the local contraction property applies at each iterate.

The base case $k = 0$ holds by assumption: $e_0 = \|\widehat{Q}^{(0)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \le r$.

Now fix $k \ge 0$ and suppose $e_k \le r$. Then $\widehat{Q}^{(k)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$, so

$$\|\mathcal{T}(\widehat{Q}^{(k)}) - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \le \rho_\star e_k.$$

Using the inexact update,

$$\widehat{Q}^{(k+1)} = \mathcal{T}(\widehat{Q}^{(k)}) + \vec{\xi}_k,$$

we obtain

$$\begin{aligned}e_{k+1} &= \|\widehat{Q}^{(k+1)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \\ &\le \|\mathcal{T}(\widehat{Q}^{(k)}) - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} + \|\vec{\xi}_k\|_{2,\mu^\star} \\ &\le \rho_\star e_k + \xi_k \;\le\; \rho_\star r + (1 - \rho_\star) r \;=\; r,\end{aligned}$$

where we used $e_k \le r$ and $\xi_k \le (1 - \rho_\star) r$ in the last step. Thus $e_{k+1} \le r$ and the induction is complete.

Step 2: Unrolling the inexact recursion. From the display above we have, for every $k \ge 0$,

$$e_{k+1} \le \rho_\star e_k + \xi_k.$$

Iterating this recursion yields, for any $k \ge 1$,

$$e_k \le \rho_\star^k e_0 + \sum_{j=0}^{k-1} \rho_\star^{k-1-j}\, \xi_j.$$

This is precisely the claimed bound, and the case $k = 0$ holds trivially. □

F.1. Regret of a Single Inexact Picard Step

F.1.1. Local maximal inequality

Let $O_1, \dots, O_n \in \mathcal{O}$ be independent random variables. For any function $f : \mathcal{O} \to \mathbb{R}$, define

$$\|f\| := \sqrt{\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[f(O_i)^2]}. \tag{12}$$

For a star-shaped class of functions $\mathcal{F}$ and a radius $\delta \in (0, \infty)$, define the localized Rademacher complexity

$$\mathcal{R}_n(\mathcal{F}, \delta) := \mathbb{E}\left[\sup_{\substack{f \in \mathcal{F} \\ \|f\| \le \delta}} \frac{1}{n}\sum_{i=1}^{n} \epsilon_i f(O_i)\right],$$

where ϵ_i are i.i.d. Rademacher random variables.

The following lemma provides a local maximal inequality and restates Lemma 11 of (Foster & Syrgkanis, 2023) (see also Lemma 11 of van der Laan et al. (2025)).

Lemma 10 (Local maximal inequality)**.** *Let $\mathcal{F}$ be a star-shaped class of functions satisfying $\sup_{f \in \mathcal{F}} \|f\|_\infty \le M$. Let $\delta = \delta_n \in (0, 1)$ satisfy the critical radius condition $\mathcal{R}_n(\mathcal{F}, \delta) \le \delta^2$, and suppose that, as $n \to \infty$,*

$$\frac{1}{\sqrt{n}}\sqrt{\log\log(1/\delta_n)} = o(\delta_n).$$

Then there exists a universal constant $C > 0$ such that, for all $\eta \in (0, 1)$, with probability at least $1 - \eta$, every $f \in \mathcal{F}$ satisfies

$$\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big) \le C\Big(\delta^2 + \delta\,\|f\|\Big) + C\Big(\frac{\sqrt{\log(1/\eta)}\,\|f\|}{\sqrt{n}} + \frac{M\,\log(1/\eta)}{n}\Big).$$

Proof. Lemma 11 in van der Laan et al. (2025) shows that there exists a universal constant $C > 0$ such that, for all $u \ge 1$, with probability at least $1 - e^{-u^2}$, every $f \in \mathcal{F}$ satisfies

$$\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big) \le C\Big(\delta^2 + \delta\,\|f\| + \frac{u\,\|f\|}{\sqrt{n}} + \frac{M\,u^2}{n}\Big).$$

Set $u := \sqrt{\log(1/\eta)}$, so that $e^{-u^2} = e^{-\log(1/\eta)} = \eta$. Substituting this choice of u into the above inequality yields, with probability at least $1 - \eta$, every $f \in \mathcal{F}$ satisfies

$$\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big) \le C\Big(\delta^2 + \delta\,\|f\|\Big) + C\Big(\frac{\sqrt{\log(1/\eta)}\,\|f\|}{\sqrt{n}} + \frac{M\,\log(1/\eta)}{n}\Big).$$

□

The following lemma bounds the localized Rademacher complexity in terms of the uniform entropy integral and is a direct consequence of Theorem 2.1 of Van Der Vaart & Wellner (2011).

For any distribution Q and any uniformly bounded function class $\mathcal{F}$, let $N(\varepsilon, \mathcal{F}, L^2(Q))$ denote the ε-covering number of $\mathcal{F}$ under the $L^2(Q)$ norm (Van Der Vaart & Wellner, 1996). Define the uniform entropy integral of $\mathcal{F}$ by

$$\mathcal{J}(\delta, \mathcal{F}) := \int_0^\delta \sup_Q \sqrt{\log N(\epsilon, \mathcal{F}, L^2(Q))}\, d\epsilon,$$

where the supremum is taken over all discrete probability distributions Q.

Lemma 11. *Let $\mathcal{F}$ be a star-shaped class of functions such that $\sup_{f \in \mathcal{F}} \|f\|_\infty \le M$. Then, for every $\delta > 0$,*

$$\mathcal{R}_n(\mathcal{F}, \delta) \lesssim \frac{1}{\sqrt{n}}\,\mathcal{J}(\delta, \mathcal{F})\left(1 + \frac{\mathcal{J}(\delta, \mathcal{F})}{\delta\sqrt{n}}\right),$$

where the implicit constant depends only on M.

Proof. This bound follows directly from the argument in the proof of Theorem 2.1 of Van Der Vaart & Wellner (2011); see in particular the step where the local Rademacher complexity is controlled by the uniform entropy integral for star-shaped classes. □

F.2. Technical lemmas

Recall the weight function d in C5, and let $\hat{d}$ denote a (possibly data–dependent) estimate of this weight.

Let $\mathcal{F}$ be a convex function class and let $\hat{Q}^{\text{init}} \in \mathcal{F}$ be an initial (possibly data–dependent) estimate. Let $\widehat{\pi}^{\text{init}} := \pi_{\hat{Q}^{\text{init}}}$ denote the softmax policy induced by $\hat{Q}^{\text{init}}$, and write

$$(\widehat{\pi}^{\text{init}} Q)(s') := \sum_{a'} \widehat{\pi}^{\text{init}}(a' \mid s')\, Q(s', a')$$

for its action–value expectation.

In what follows, expectations are taken with the nuisance estimators held fixed. Define the stationary-weighted population risk

$$\hat{R}_0(Q) := \mathbb{E}_{\nu_b}\Big[d(S, A)\,\big\{R + \gamma(\widehat{\pi}^{\text{init}}\hat{Q}^{\text{init}})(S') - Q(S, A)\big\}^2\Big].$$

Define the empirical minimizer

$$\hat{Q}_n := \arg\min_{Q \in \mathcal{F}} \hat{R}_n(Q),$$

where

$$\hat{R}_n(Q) := \frac{1}{n}\sum_{i=1}^{n} \hat{d}(S_i, A_i)\,\big\{R_i + \gamma(\widehat{\pi}^{\text{init}}\hat{Q}^{\text{init}})(S_i') - Q(S_i, A_i)\big\}^2.$$

Lemma 12 shows that this stationary–reweighted population objective is correctly specified for the projected soft Bellman operator $\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})$ under Condition C5.

Lemma 12 (Fixed-point operator as population risk minimizer)**.** *Under Condition C5,*

$$\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) = \arg\min_{Q\in\overline{\mathcal{F}}} \hat{R}_0(Q).$$

Proof. Consider the stationary–population objective

$$\tilde{R}_0(Q) := \mathbb{E}_{\mu^\star}\left[\left\{R+\gamma(\widehat{\pi}^{\text{init}}\hat{Q}^{\text{init}})(S') - Q(S,A)\right\}^2 \;\Big|\right],$$

where $\mu^\star$ is the stationary state–action distribution used to define the $L^2(\mu^\star)$ norm in the main text. By definition of the projected Bellman operator,

$$\mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}}) = \arg\min_{Q\in\overline{\mathcal{F}}} \tilde{R}_0(Q),$$

since this is precisely the $L^2(\mu^\star)$ projection of the soft Bellman target

$$g(S,A) := R + \gamma(\widehat{\pi}^{\text{init}}\hat{Q}^{\text{init}})(S')$$

onto $\overline{\mathcal{F}}$.

Condition C5 guarantees that, for functions in $\mathcal{F}$, reweighting by d transfers the $L^2(\mu^\star)$ inner product to the behavior distribution ν_b. In particular, for every $Q \in \mathcal{F}$ and direction $h \in \mathcal{F}$,

$$\begin{aligned} D\tilde{R}_0(Q)[h] &= -2\,\mathbb{E}_{\mu^\star}\left[h(S,A)\left\{g(S,A)-Q(S,A)\right\}\right] \\ &= -2\,\mathbb{E}_{\nu_b}\left[d(S,A)\,h(S,A)\left\{(g-Q)(S,A)\right\}\right] \\ &= D\hat{R}_0(Q)[h], \end{aligned}$$

where the second equality follows from the restricted stationary overlap identity in C5 applied to the product $h(S,A)\{g(S,A)-Q(S,A)\}$.

In particular, at the minimizer $Q^\dagger := \mathcal{T}_{\mathcal{F}}(\hat{Q}^{\text{init}})$ of $\tilde{R}_0$ over $\overline{\mathcal{F}}$, the first–order optimality conditions give $D\tilde{R}_0(Q^\dagger)[h] = 0$ for all $h \in \mathcal{F}$, and hence also $D\hat{R}_0(Q^\dagger)[h] = 0$ for all $h \in \mathcal{F}$. Since $\hat{R}_0$ is a strictly convex quadratic functional in Q, these conditions uniquely characterize its minimizer over $\overline{\mathcal{F}}$, so $Q^\dagger = \arg\min_{Q\in\overline{\mathcal{F}}} \hat{R}_0(Q)$. □

Lemma 13 (Empirical-process bound for one weighted regression)**.** *Assume Conditions C7 and C9. Let P_0 denote the law of (S,A,R,S') under ν_b and dynamics P, and let P_n be the empirical measure. Let w be any state–action function uniformly bounded by M.*

For each k, define the softmax policy induced by the kth iterate as $\widehat{\pi}^{(k)} := \pi_{\widehat{Q}^{(k)}}$, and write

$$(\widehat{\pi}^{(k)}Q)(s') := \sum_{a'} \widehat{\pi}^{(k)}(a' \mid s')\, Q(s',a').$$

Then there exists a constant $C = C(M) > 0$ such that, for all $\eta \in (0,1)$, with probability at least $1-\eta$,

$$\begin{aligned} &\Big|(P_n - P_0)\big[w\left(\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\right) \\ &\qquad\qquad \times \left\{R + \gamma(\widehat{\pi}^{(k)}\,\widehat{Q}^{(k)})(S') - \widehat{Q}^{(k+1)}(S,A)\right\}\big]\Big| \\ &\le\; C\left(\delta_n^2 + \delta_n \left\|w\big(\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\big)\right\|_{L^2(P_0)}\right) \\ &\quad + \; C\left(\frac{\log(1/\eta)\left\|w\big(\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\big)\right\|_{L^2(P_0)}}{\sqrt{n}}\right) \\ &\quad + \; C\,\frac{\log(1/\eta)}{n}. \end{aligned}$$

Proof of Lemma 13. Fix a bounded weight function w with $\|w\|_\infty \le M$. Define the function class

$$\begin{aligned} \mathcal{G}_w := \Big\{ g_{Q_1,Q_2,V_1} : (s,a,r,s') \mapsto w(s,a)\left(Q_1(s,a) - Q_2(s,a)\right) \\ \times \left\{r + \gamma V_1(s') - Q_2(s,a)\right\} :\; Q_1,Q_2,V_1 \in \mathcal{F}_{\text{ext}} \Big\}. \end{aligned}$$

By Condition C7, every $Q \in \mathcal{F}_{\text{ext}}$ and R are uniformly bounded by M. Thus, for any $g_{Q_1,Q_2,V_1} \in \mathcal{G}_w$,

$$\begin{aligned} |g_{Q_1,Q_2,V_1}(s,a,r,s')| &\le \|w\|_\infty \left|Q_1(s,a) - Q_2(s,a)\right| \\ &\qquad \times \left(|r| + \gamma|V_1(s')| + |Q_2(s,a)|\right) \\ &\le C_0(M)\left|Q_1(s,a) - Q_2(s,a)\right|. \end{aligned}$$

for some constant $C_0(M) < \infty$ depending only on M. In particular, $\mathcal{G}_w$ is uniformly bounded, and

$$\|g_{Q_1,Q_2,V_1}\|_{L^2(P_0)} \;\le\; C_0(M)\,\|w\,(Q_1 - Q_2)\|_{L^2(P_0)}. \tag{13}$$

Entropy control. By Condition C9, the uniform entropy integral $\mathcal{J}(\delta,\mathcal{F}_{\text{ext}})$ is finite and determines the critical radius δ_n. The map $(Q_1,Q_2,V_1) \mapsto g_{Q_1,Q_2,V_1}$ is a Lipschitz (in sup-norm) multilinear transformation of (Q_1,Q_2,V_1), with Lipschitz constant depending only on M and γ, and multiplied only by uniformly bounded factors (w, R, and Q_2). By standard permanence properties of entropy under Lipschitz maps and bounded multipliers (e.g., Thm. 2.6.18 in (Van Der Vaart & Wellner, 1996)), the local entropy of $\mathcal{G}_w$ is therefore controlled by that of $\mathcal{F}_{\text{ext}}$: there exists $C_1 = C_1(M)$ such that

$$\mathcal{J}(\delta,\mathcal{G}_w) \;\le\; C_1\,\mathcal{J}(\delta,\mathcal{F}_{\text{ext}}), \qquad \forall\,\delta > 0.$$

Hence $\mathcal{G}_w$ has the same critical radius as $\mathcal{F}_{\text{ext}}$, up to a multiplicative constant (absorbed into $C(M)$). We therefore continue to denote this radius by δ_n.

Finally, we replace $\mathcal{G}_w$ by its star-shaped hull

$$\mathcal{G}_w^\circ := \{\, tg : g \in \mathcal{G}_w,\; t \in [0,1] \,\}.$$

Standard permanence properties of uniform entropy numbers (e.g., Thm. 2.6.18 in (Van Der Vaart & Wellner, 1996)) guarantee that $\mathcal{J}(\delta, \mathcal{G}_w^\circ)$ is bounded, up to a universal constant factor, by $\mathcal{J}(\delta, \mathcal{G}_w)$. Hence $\mathcal{G}_w^\circ$ has the same critical radius δ_n, up to constants absorbed into $C(M)$. Because the localized maximal inequality applies to star-shaped classes, we work henceforth with $\mathcal{G}_w^\circ$ (without changing any resulting bounds).

Local empirical-process bound. The class $\mathcal{G}_w^\circ$ is uniformly bounded, star-shaped, and—by the entropy bound above—has localized entropy controlled by that of $\mathcal{F}_{\text{ext}}$. Lemma 11 therefore yields $\mathcal{R}_n(\mathcal{G}_w^\circ, \delta_n) \lesssim \delta_n^2$. Since δ_n satisfies the critical-radius condition, Lemma 10 applies to $\mathcal{G}_w^\circ$ and gives: for all $\eta \in (0,1)$, with probability at least $1-\eta$, every $g \in \mathcal{G}_w^\circ$ satisfies

$$|(P_n - P_0)g| \;\le\; C\big(\delta_n^2 + \delta_n \|g\|_{L^2(P_0)}\big) + C\left(\frac{\sqrt{\log(1/\eta)}\,\|g\|_{L^2(P_0)}}{\sqrt{n}} + \frac{\log(1/\eta)}{n}\right),$$

for some constant $C = C(M)$.

Specialization to the inexact regression function. For iteration k, set

$$Q_1 = \mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}), \qquad Q_2 = \widehat{Q}^{(k+1)}, \qquad V_1 = (\widehat{\pi}^{(k)}\widehat{Q}^{(k)}),$$

so $Q_1, Q_2, V_1 \in \mathcal{F}_{\text{ext}}$. The associated regression function $g_k(s,a,r,s')$

$$g_k(s,a,r,s') = w(s,a)\,(Q_1 - Q_2)(s,a)\,\{r + \gamma V_1(s') - Q_2(s,a)\}$$

belongs to $\mathcal{G}_w$. By (13) and boundedness of w, R, and $\mathcal{F}_{\text{ext}}$,

$$\|g_k\|_{L^2(P_0)} \;\le\; C_0(M)\,\big\|w\,\big(\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\big)\big\|_{L^2(P_0)}.$$

Applying the uniform deviation bound to g_k and absorbing constants into $C = C(M)$ gives, with probability at least $1-\eta$,

$$\begin{aligned}
&\Big|(P_n - P_0)\big[w\,\big(\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\big) \times \big\{R + \gamma\big(\widehat{\pi}^{(k)}\widehat{Q}^{(k)}\big)(S') - \widehat{Q}^{(k+1)}(S,A)\big\}\big]\Big| \\
&\le C\Big(\delta_n^2 + \delta_n \big\|w\big(\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\big)\big\|_{L^2(P_0)}\Big) \\
&\quad + C\,\frac{\sqrt{\log(1/\eta)}\,\big\|w\big(\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\big)\big\|_{L^2(P_0)}}{\sqrt{n}} \\
&\quad + C\,\frac{\log(1/\eta)}{n}.
\end{aligned}$$

which is the desired bound. □

F.3. Regret for a Single Inexact Picard Iteration

Proof of Lemma 5. Fix $0 \le k \le K-1$ and abbreviate

$$\widehat{d}^{(k)} := \widehat{d}^{(k)}, \quad w := d, \quad \widehat{Q} := \widehat{Q}^{(k)}, \quad \widehat{Q}^+ := \widehat{Q}^{(k+1)}.$$

Let P_0 denote the law of (S, A, R, S') under ν_b and P_n the empirical measure on $\mathcal{D}_n$.

Step 1: Empirical first-order optimality. By construction, $\widehat{Q}^+$ is the (weighted) empirical risk minimizer:

$$\widehat{Q}^+ = \arg\min_{Q \in \mathcal{F}} \frac{1}{n}\sum_{i=1}^n \widehat{d}^{(k)}(S_i, A_i)\,\big\{R_i + \gamma(\widehat{\pi}^{(k)}\widehat{Q})(S_i') - Q(S_i, A_i)\big\}^2.$$

By convexity of $\mathcal{F}$ and first-order optimality, for all $Q \in \mathcal{F}$,

$$\frac{1}{n}\sum_{i=1}^n \widehat{d}^{(k)}(S_i, A_i)\,\big\{Q(S_i, A_i) - \widehat{Q}^+(S_i, A_i)\big\} \times \big\{R_i + \gamma(\widehat{\pi}^{(k)}\widehat{Q})(S_i') - \widehat{Q}^+(S_i, A_i)\big\} \;\le\; 0.$$

By Lemma 12, the projected Bellman update $\mathcal{T}_{\mathcal{F}}(\widehat{Q})$ belongs to $\mathcal{F}$. Taking $Q = \mathcal{T}_{\mathcal{F}}(\widehat{Q})$ and rearranging, we obtain

$$\begin{aligned}
&P_0\big[\widehat{d}^{(k)}\,\Delta_k\,\{R + \gamma(\widehat{\pi}^{(k)}\widehat{Q})(S') - \widehat{Q}^+(S,A)\}\big] \\
&\le\; (P_0 - P_n)\big[\widehat{d}^{(k)}\,\Delta_k\,\{R + \gamma(\widehat{\pi}^{(k)}\widehat{Q})(S') - \widehat{Q}^+(S,A)\}\big],
\end{aligned} \tag{14}$$

where we set

$$\Delta_k := \mathcal{T}_{\mathcal{F}}(\widehat{Q}) - \widehat{Q}^+.$$

Step 2: Population quadratic lower bound. Using the Bellman target

$$\mathcal{T}(\widehat{Q})(S,A) = \mathbb{E}\big[R + \gamma(\widehat{\pi}^{(k)}\widehat{Q})(S') \mid S, A\big],$$

the law of total expectation yields

$$\begin{aligned}
P_0\big[\widehat{d}^{(k)}\,\Delta_k\,\{R + \gamma(\widehat{\pi}^{(k)}\widehat{Q})(S') - \widehat{Q}^+\}\big] &= P_0\big[\widehat{d}^{(k)}\,\Delta_k\,\{\mathcal{T}(\widehat{Q}) - \widehat{Q}^+\}\big].
\end{aligned}$$

Add and subtract w:

$$\begin{aligned}
&P_0\big[\widehat{d}^{(k)}\,\Delta_k\,\{\mathcal{T}(\widehat{Q}) - \widehat{Q}^+\}\big] \\
&= P_0\big[w\,\Delta_k\,\{\mathcal{T}(\widehat{Q}) - \widehat{Q}^+\}\big] \\
&\quad + P_0\big[(\widehat{d}^{(k)} - w)\,\Delta_k\,\{\mathcal{T}(\widehat{Q}) - \widehat{Q}^+\}\big].
\end{aligned}$$

For the first term, use the population optimality of $\mathcal{T}_{\mathcal{F}}(\widehat{Q})$ under the weighted risk $\hat{R}_0$ in Lemma 12. Writing

$$\mathcal{T}(\widehat{Q}) - \widehat{Q}^+ = \{\mathcal{T}(\widehat{Q}) - \mathcal{T}_{\mathcal{F}}(\widehat{Q})\} + \Delta_k,$$

we have

$$\begin{aligned}
&P_0\big[w\,\Delta_k\,\{\mathcal{T}(\widehat{Q}) - \widehat{Q}^+\}\big] \\
&= P_0\big[w\,\Delta_k^2\big] + P_0\big[w\,\Delta_k\,\{\mathcal{T}(\widehat{Q}) - \mathcal{T}_{\mathcal{F}}(\widehat{Q})\}\big].
\end{aligned}$$

By Lemma 12, the second term is nonnegative (it is the directional derivative of the population risk at its minimizer), so

$$P_0\big[w\,\Delta_k\,\{\mathcal{T}(\widehat{Q})-\widehat{Q}^+\}\big] \;\geq\; P_0[w\,\Delta_k^2] = \|\Delta_k\|_{2,w\nu_b}^2.$$

Recall that

$$\|\Delta_k\|_{2,w\nu_b}^2 = P_0\big[w\,\Delta_k^2\big], \qquad \|f\|_{2,w\nu_b}^2 := P_0[wf^2].$$

Using the decomposition

$$\mathcal{T}(\widehat{Q})-\widehat{Q}^+ = \{\mathcal{T}(\widehat{Q})-\mathcal{T}_{\mathcal{F}}(\widehat{Q})\}+\Delta_k$$

and the population optimality of $\mathcal{T}_{\mathcal{F}}(\widehat{Q})$ from Lemma 12, we have

$$\begin{aligned}
&P_0\big[w\,\Delta_k\,\{\mathcal{T}(\widehat{Q})-\widehat{Q}^+\}\big]\\
&= P_0\big[w\,\Delta_k^2\big] + P_0\big[w\,\Delta_k\,\{\mathcal{T}(\widehat{Q})-\mathcal{T}_{\mathcal{F}}(\widehat{Q})\}\big]\\
&\geq P_0\big[w\,\Delta_k^2\big] = \|\Delta_k\|_{2,w\nu_b}^2.
\end{aligned}$$

For the second term in the decomposition of $P_0\big[\widehat{d}^{(k)}\,\Delta_k\,\{\mathcal{T}(\widehat{Q})-\widehat{Q}^+\}\big]$, apply Cauchy–Schwarz:

$$\begin{aligned}
&P_0\big[(\widehat{d}^{(k)}-w)\,\Delta_k\,\{\mathcal{T}(\widehat{Q})-\widehat{Q}^+\}\big]\\
&= P_0\big[w\Big(\tfrac{\widehat{d}^{(k)}}{w}-1\Big)\,\Delta_k\,\{\mathcal{T}(\widehat{Q})-\widehat{Q}^+\}\big]\\
&\geq -\Big\|\big(\tfrac{\widehat{d}^{(k)}}{w}-1\big)\,\{\mathcal{T}(\widehat{Q})-\widehat{Q}^+\}\Big\|_{2,w\nu_b}\,\|\Delta_k\|_{2,w\nu_b}.
\end{aligned}$$

Combining both pieces, we obtain the population lower bound

$$\begin{aligned}
&P_0\big[\widehat{d}^{(k)}\,\Delta_k\,\{R+\gamma(\widehat{\pi}^{(k)}\widehat{Q})(S')-\widehat{Q}^+\}\big] \qquad (15)\\
&\geq \|\Delta_k\|_{2,w\nu_b}^2 - \Big\|\big(\tfrac{\widehat{d}^{(k)}}{w}-1\big)\,\{\mathcal{T}(\widehat{Q})-\widehat{Q}^+\}\Big\|_{2,w\nu_b}\,\|\Delta_k\|_{2,w\nu_b}.
\end{aligned}$$

Step 3: Combine with empirical-process deviation (behavior norm). Plugging (15) into (14) and rearranging gives

$$\begin{aligned}
\|\Delta_k\|_{2,w\nu_b}^2 \leq\; &(P_0-P_n)\big[\widehat{d}^{(k)}\,\Delta_k\,\{R+\gamma(\widehat{\pi}^{(k)}\widehat{Q})(S')-\widehat{Q}^+\}\big] \qquad (16)\\
&+\Big\|\big(\tfrac{\widehat{d}^{(k)}}{w}-1\big)\,\{\mathcal{T}(\widehat{Q})-\widehat{Q}^+\}\Big\|_{2,w\nu_b}\,\|\Delta_k\|_{2,w\nu_b}.
\end{aligned}$$

We now bound the empirical-process term on the right-hand side using Lemma 13 twice: once with $w=w$ and once with $w=\widehat{d}^{(k)}-w$, both uniformly bounded by Condition C7. By Condition C8, the weights are independent of $\mathcal{D}_n$, so Lemma 13 applies conditional on the weights.

By boundedness of w,

$$\|w\,\Delta_k\|_{L^2(P_0)}^2 = P_0[w^2\Delta_k^2] \;\lesssim\; P_0[w\Delta_k^2] = \|\Delta_k\|_{2,w\nu_b}^2,$$

and similarly

$$\|(\widehat{d}^{(k)}-w)\Delta_k\|_{L^2(P_0)} \;\lesssim\; \|\Delta_k\|_{2,w\nu_b}.$$

Applying Lemma 13 with confidence level η/K and a union bound over $k=0,\dots,K-1$, we obtain that there exists $C=C(M)>0$ such that, with probability at least $1-\eta$, for all k,

$$\begin{aligned}
&\Big|(P_0-P_n)\big[\widehat{d}^{(k)}\,\Delta_k\,\{R+\gamma(\widehat{\pi}^{(k)}\widehat{Q})(S')-\widehat{Q}^+\}\big]\Big|\\
&\leq C\Big(\delta_n^2+\delta_n\,\|\Delta_k\|_{2,w\nu_b}\Big) + C\Big(\tfrac{\sqrt{\log(K/\eta)}}{\sqrt{n}}\,\|\Delta_k\|_{2,w\nu_b}+\tfrac{\log(K/\eta)}{n}\Big).
\end{aligned}$$

By definition of $\delta_{\mathrm{stat}}(n,\eta,K)$,

$$\delta_n^2+\tfrac{\log(K/\eta)}{n} \;\lesssim\; \delta_{\mathrm{stat}}(n,\eta,K)^2, \qquad \delta_n+\tfrac{\sqrt{\log(K/\eta)}}{\sqrt{n}} \;\lesssim\; \delta_{\mathrm{stat}}(n,\eta,K),$$

so, absorbing constants into C,

$$\begin{aligned}
&\Big|(P_0-P_n)\big[\widehat{d}^{(k)}\,\Delta_k\,\{R+\gamma(\widehat{\pi}^{(k)}\widehat{Q})(S')-\widehat{Q}^+\}\big]\Big| \qquad (17)\\
&\quad\leq C\Big(\delta_{\mathrm{stat}}(n,\eta,K)^2+\delta_{\mathrm{stat}}(n,\eta,K)\,\|\Delta_k\|_{2,w\nu_b}\Big).
\end{aligned}$$

Step 4: Control of the weight–error term. We now bound the weight–error contribution in (15):

$$\Big\|\Big(\tfrac{\widehat{d}^{(k)}}{w}-1\Big)\,\{\mathcal{T}(\widehat{Q})-\widehat{Q}^+\}\Big\|_{2,w\nu_b}.$$

Because this quantity is expressed in the $L^2(w\nu_b)$ norm, we first reduce it to an equivalent $L^2(\mu^\star)$ bound. By Condition C6,

$$\|f\|_{2,w\nu_b} \;\leq\; c_1\,\|f\|_{2,\mu^\star}, \quad \text{for all } f\in L^2(\mu^\star),$$

so it suffices to control

$$\Big\|\Big(\tfrac{\widehat{d}^{(k)}}{w}-1\Big)\,\{\mathcal{T}(\widehat{Q})-\widehat{Q}^+\}\Big\|_{2,\mu^\star},$$

with the understanding that the corresponding $L^2(w\nu_b)$ bound differs only by the multiplicative constant c_1.

By the triangle inequality,

$$\mathcal{T}(\widehat{Q})-\widehat{Q}^+ = \big\{\mathcal{T}(\widehat{Q})-\mathcal{T}_{\mathcal{F}}(\widehat{Q})\big\}+\Delta_k,$$

so

$$\begin{aligned}
&\Big\|\Big(\tfrac{\widehat{d}^{(k)}}{w}-1\Big)\,\{\mathcal{T}(\widehat{Q})-\widehat{Q}^+\}\Big\|_{2,\mu^\star}\\
&\leq\; \Big\|\Big(\tfrac{\widehat{d}^{(k)}}{w}-1\Big)\,\Delta_k\Big\|_{2,\mu^\star} + \Big\|\Big(\tfrac{\widehat{d}^{(k)}}{w}-1\Big)\,\{\mathcal{T}(\widehat{Q})-\mathcal{T}_{\mathcal{F}}(\widehat{Q})\}\Big\|_{2,\mu^\star}.
\end{aligned}$$

For the second term, apply Hölder's inequality and the definition of $\varepsilon_{\mathrm{Bell}}(k)$:

$$\begin{aligned}
&\Big\|\Big(\tfrac{\widehat{d}^{(k)}}{w}-1\Big)\,\{\mathcal{T}(\widehat{Q})-\mathcal{T}_{\mathcal{F}}(\widehat{Q})\}\Big\|_{2,\mu^\star}\\
&\leq \Big\|\tfrac{\widehat{d}^{(k)}}{w}-1\Big\|_{2,\mu^\star}\,\|\mathcal{T}(\widehat{Q})-\mathcal{T}_{\mathcal{F}}(\widehat{Q})\|_\infty\\
&\lesssim \varepsilon_{\mathrm{wt}}(k)\,\varepsilon_{\mathrm{Bell}}(k),
\end{aligned}$$

where we used Condition C7 to bound the sup–norm of the Bellman residual by a constant multiple of its $L^2(\mu^\star)$ norm.

For the first term, we again use Hölder's inequality:

$$\left\|\left(\frac{\widehat{d}^{(k)}}{w}-1\right)\Delta_k\right\|_{2,\mu^\star} \le \left\|\frac{\widehat{d}^{(k)}}{w}-1\right\|_{2,\mu^\star}\|\Delta_k\|_\infty = \varepsilon_{\mathrm{wt}}(k)\,\|\Delta_k\|_\infty.$$

To control $\|\Delta_k\|_\infty$, apply the Hölder-type L^∞–L^2 relation (Condition C3) with $Q_1 = \mathcal{T}(\widehat{Q})$ and $Q_2 = \widehat{Q}^+$:

$$\|\Delta_k\|_\infty = \|\mathcal{T}(\widehat{Q}) - \widehat{Q}^+\|_\infty \;\le\; C_\infty\,\|\mathcal{T}(\widehat{Q}) - \widehat{Q}^+\|_{2,\mu^\star}^{\alpha}.$$

By the triangle inequality and the definition of $\varepsilon_{\mathrm{Bell}}(k)$,

$$\|\mathcal{T}(\widehat{Q}) - \widehat{Q}^+\|_{2,\mu^\star} \le \left\|\mathcal{T}(\widehat{Q}) - \mathcal{T}_{\mathcal{F}}(\widehat{Q})\right\|_{2,\mu^\star} + \|\Delta_k\|_{2,\mu^\star} \le \varepsilon_{\mathrm{Bell}}(k) + \|\Delta_k\|_{2,\mu^\star}.$$

Combining these bounds,

$$\|\Delta_k\|_\infty \;\le\; C_\infty\left(\|\Delta_k\|_{2,\mu^\star} + \varepsilon_{\mathrm{Bell}}(k)\right)^{\alpha}.$$

Hence

$$\left\|\left(\frac{\widehat{d}^{(k)}}{w}-1\right)\Delta_k\right\|_{2,\mu^\star} \le \varepsilon_{\mathrm{wt}}(k)\,\|\Delta_k\|_\infty \le C_\infty\varepsilon_{\mathrm{wt}}(k)\left(\|\Delta_k\|_{2,\mu^\star} + \varepsilon_{\mathrm{Bell}}(k)\right)^{\alpha}.$$

Putting the two pieces together, we obtain

$$\left\|\left(\frac{\widehat{d}^{(k)}}{w}-1\right)\left\{\mathcal{T}(\widehat{Q}) - \widehat{Q}^+\right\}\right\|_{2,\mu^\star} \tag{18}$$

$$\lesssim\; \varepsilon_{\mathrm{wt}}(k)\,\varepsilon_{\mathrm{Bell}}(k) + \varepsilon_{\mathrm{wt}}(k)\left(\|\Delta_k\|_{2,\mu^\star} + \varepsilon_{\mathrm{Bell}}(k)\right)^{\alpha}, \tag{19}$$

where the implicit constant depends only on C_∞ and the bounds in Condition C7.

Step 5: Solve in the behavior norm, then translate to $\mu^\star$. Combining the basic inequality (16), the empirical-process bound (17), and the weight–error bound (18), we obtain: with probability at least $1-\eta$,

$$\|\Delta_k\|_{2,w\nu_b}^2 \le C\Big(\delta_{\mathrm{stat}}(n,\eta,K)^2 + \delta_{\mathrm{stat}}(n,\eta,K)\,\|\Delta_k\|_{2,w\nu_b}\Big) + C\,\varepsilon_{\mathrm{wt}}(k)\,\varepsilon_{\mathrm{Bell}}(k)\,\|\Delta_k\|_{2,w\nu_b} + C\,\varepsilon_{\mathrm{wt}}(k)\left(\|\Delta_k\|_{2,w\nu_b} + \varepsilon_{\mathrm{Bell}}(k)\right)^{\alpha}\|\Delta_k\|_{2,w\nu_b}, \tag{20}$$

for some constant $C = C(M) > 0$.

By Condition C6, we have

$$c_1^{-1}\,\|\Delta_k\|_{2,w\nu_b} \;\le\; \|\Delta_k\|_{2,\mu^\star} \;\le\; c_0\,\|\Delta_k\|_{2,w\nu_b},$$

for constants $c_0, c_1 \in (0,\infty)$ depending only on the bounds on w. Applying these equivalences to (20) and absorbing constants into C, we can rewrite the inequality in terms of $x_k := \|\Delta_k\|_{2,\mu^\star}$ as

$$x_k^2 \le C\Big(\delta_{\mathrm{stat}}(n,\eta,K)^2 + \delta_{\mathrm{stat}}(n,\eta,K)\,x_k\Big) + C\,\varepsilon_{\mathrm{wt}}(k)\,\varepsilon_{\mathrm{Bell}}(k)\,x_k + C\,\varepsilon_{\mathrm{wt}}(k)\left(x_k + \varepsilon_{\mathrm{Bell}}(k)\right)^{\alpha} x_k.$$

A standard one-dimensional calculus / Young-inequality argument applied to this scalar inequality then yields

$$\|\Delta_k\|_{2,\mu^\star} \le C\Big(\delta_{\mathrm{stat}}(n,\eta,K) + \varepsilon_{\mathrm{wt}}(k)^{\frac{1}{1-\alpha}} + \varepsilon_{\mathrm{wt}}(k)\,\varepsilon_{\mathrm{Bell}}(k)\Big)$$

for a (possibly larger) constant $C = C(c_0, c_1, M) > 0$, which is the claimed bound.

□

G. Proof of main convergence bounds

G.1. Technical lemma

Lemma 14. *Assume C1–C3. Let $r \in (0, r_{\max})$ and suppose $\widehat{Q}^{(k)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$. Then,*

$$\varepsilon_{\mathrm{Bell}}(k) \le (1 + 2\,\rho_{\mathrm{eff}}(r))\,\varepsilon_{\mathcal{F}} \;+\; 2\,\rho_{\mathrm{eff}}(r)\,\|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star}.$$

Proof. By the triangle inequality,

$$\begin{aligned}\varepsilon_{\mathrm{Bell}}(k) &= \left\|\mathcal{T}(\widehat{Q}^{(k)}) - \mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)})\right\|_{2,\mu^\star}\\ &\le \left\|\mathcal{T}(\widehat{Q}^{(k)}) - \mathcal{T}(Q^\star)\right\|_{2,\mu^\star} + \left\|\mathcal{T}(Q^\star) - \mathcal{T}_{\mathcal{F}}(Q^\star)\right\|_{2,\mu^\star}\\ &\quad + \left\|\mathcal{T}_{\mathcal{F}}(Q^\star) - \mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)})\right\|_{2,\mu^\star}\\ &\le \varepsilon_{\mathcal{F}} \;+\; 2\left\|\mathcal{T}(\widehat{Q}^{(k)}) - \mathcal{T}(Q^\star)\right\|_{2,\mu^\star}.\end{aligned}$$

By Theorem 1, the unprojected operator $\mathcal{T}$ is a $\rho_{\mathrm{eff}}(r)$–contraction on $\mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$, we have

$$\left\|\mathcal{T}(\widehat{Q}^{(k)}) - \mathcal{T}(Q^\star)\right\|_{2,\mu^\star} \;\le\; \rho_{\mathrm{eff}}(r)\,\|\widehat{Q}^{(k)} - Q^\star\|_{2,\mu^\star},$$

and hence

$$\varepsilon_{\mathrm{Bell}}(k) \le \varepsilon_{\mathcal{F}} \;+\; 2\,\rho_{\mathrm{eff}}(r)\,\|\widehat{Q}^{(k)} - Q^\star\|_{2,\mu^\star}.$$

Applying the triangle inequality once more,

$$\begin{aligned}\|\widehat{Q}^{(k)} - Q^\star\|_{2,\mu^\star} &\le \|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \;+\; \|Q^\star_{\mathcal{F}} - Q^\star\|_{2,\mu^\star}\\ &\le \|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \;+\; \varepsilon_{\mathcal{F}},\end{aligned}$$

so

$$\begin{aligned}\varepsilon_{\mathrm{Bell}}(k) &\le \varepsilon_{\mathcal{F}} \;+\; 2\,\rho_{\mathrm{eff}}(r)\left(\|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} + \varepsilon_{\mathcal{F}}\right)\\ &= (1 + 2\,\rho_{\mathrm{eff}}(r))\,\varepsilon_{\mathcal{F}} \;+\; 2\,\rho_{\mathrm{eff}}(r)\,\|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star}.\end{aligned}$$

□

G.2. Proof of Theorem 3

Proof of Theorem 3. We proceed by induction. We will show that $\widehat{Q}^{(k)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$ for all $0 \le k \le K$. This holds for $k = 0$ by assumption C10. Throughout the proof, whenever we state that a bound holds with high probability, we mean that it holds uniformly for all $0 \le k \le K$.

Fix an iteration $0 \le k \le K-1$ and suppose $\widehat{Q}^{(k)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$. Then

$$\|\widehat{Q}^{(k+1)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \le \|\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} + \|\widehat{Q}^{(k+1)} - \mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)})\|_{2,\mu^\star}. \tag{21}$$

By the local contraction property in Theorem 2 and the fixed-point identity $Q^\star_{\mathcal{F}} = \mathcal{T}_{\mathcal{F}}(Q^\star_{\mathcal{F}})$, the first term satisfies

$$\|\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \le \|\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \mathcal{T}_{\mathcal{F}}(Q^\star_{\mathcal{F}})\|_{2,\mu^\star} \le \rho_{\text{eff}}(r)\, \|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star}. \tag{22}$$

Fix $\eta \in (0,1)$. By Lemma 5, for all $0 \le k \le K-1$, with probability at least $1-\eta$, the second term satisfies

$$\left\|\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\right\|_{2,\mu^\star} \le C\,\delta_{\text{stat}}(n,\eta,K) + C\,\varepsilon_{\text{wt}}(k)^{\frac{1}{1-\alpha}} + C\varepsilon_{\text{wt}}(k)\,\varepsilon_{\text{Bell}}(k).$$

Since $\widehat{Q}^{(k)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$ by the induction hypothesis, Lemma 14 yields

$$\varepsilon_{\text{Bell}}(k) \le \big(1 + 2\rho_{\text{eff}}(r)\big)\,\varepsilon_{\mathcal{F}} \;+\; 2\rho_{\text{eff}}(r)\,\|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star}.$$

Hence,

$$\begin{aligned}&\left\|\mathcal{T}_{\mathcal{F}}(\widehat{Q}^{(k)}) - \widehat{Q}^{(k+1)}\right\|_{2,\mu^\star}\\ &\le C\,\delta_{\text{stat}}(n,\eta,K) \;+\; C\,\varepsilon_{\text{wt}}(k)^{\frac{1}{1-\alpha}} \\ &\quad + C\varepsilon_{\text{wt}}(k)\Big[\big(1+2\rho_{\text{eff}}(r)\big)\,\varepsilon_{\mathcal{F}} + 2\rho_{\text{eff}}(r)\,e_k\Big],\end{aligned} \tag{23}$$

where $e_k := \|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star}$.

Plugging (22) and (23) into (21), we obtain

$$\begin{aligned}\|\widehat{Q}^{(k+1)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} &\le \rho_{\text{eff}}(r)\,\|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \\ &\quad + C\,\delta_{\text{stat}}(n,\eta,K) + C\,\varepsilon_{\text{wt}}(k)^{\frac{1}{1-\alpha}} \\ &\quad + C\,\varepsilon_{\text{wt}}(k)\Big[\big(1+2\rho_{\text{eff}}(r)\big)\,\varepsilon_{\mathcal{F}} + 2\rho_{\text{eff}}(r)\,e_k\Big].\end{aligned} \tag{24}$$

Collecting the terms multiplying $\|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star}$ and the residual terms, and enlarging C if necessary (using $1 + 2\rho_{\text{eff}}(r) \le 3$), this becomes

$$\begin{aligned}&\|\widehat{Q}^{(k+1)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star}\\ &\quad\le \rho_{\text{eff}}(r)\big(1 + 2C\varepsilon_{\text{wt}}(k)\big)\,\|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star} \\ &\qquad + C\,\delta_{\text{stat}}(n,\eta,K) + C\,\varepsilon_{\text{wt}}(k)^{\frac{1}{1-\alpha}} \\ &\qquad + C\,\varepsilon_{\text{wt}}(k)\,\varepsilon_{\mathcal{F}}.\end{aligned}$$

Define

$$e_k := \|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star},$$

so the recursion reads, for a (possibly larger) constant $C > 0$,

$$\begin{aligned}e_{k+1} &\le \rho_{\text{eff}}(r)\big(1 + 2C\,\varepsilon_{\text{wt}}(k)\big)\,e_k \\ &\quad + C\,\delta_{\text{stat}}(n,\eta,K) + C\,\varepsilon_{\text{wt}}(k)^{\frac{1}{1-\alpha}} + C\,\varepsilon_{\text{wt}}(k)\,\varepsilon_{\mathcal{F}}.\end{aligned}$$

By assumption C10, there exists a radius $r \in (0, r_{\max})$ such that for all $0 \le k \le K-1$,

$$\rho_{\text{eff}}(r)\,\big(1 + 2C\,\varepsilon_{\text{wt}}(k)\big) < 1,$$

$$C\,\delta_{\text{stat}}(n,\eta,K) + C\,\varepsilon_{\text{wt}}(k)^{\frac{1}{1-\alpha}} + C\,\varepsilon_{\text{wt}}(k)\,\varepsilon_{\mathcal{F}} \;\le\; \Big[1 - \rho_{\text{eff}}(r)\,\big(1 + 2C\,\varepsilon_{\text{wt}}(k)\big)\Big]\,r.$$

Hence, whenever $e_k \le r$,

$$\begin{aligned}e_{k+1} &\le \rho_{\text{eff}}(r)\,\big(1 + 2C\,\varepsilon_{\text{wt}}(k)\big)\,e_k \\ &\quad + \big[1 - \rho_{\text{eff}}(r)\,\big(1 + 2C\,\varepsilon_{\text{wt}}(k)\big)\big]\,r \\ &\le r.\end{aligned}$$

Thus, by induction, $\widehat{Q}^{(k)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$ for all $k \le K$.

We now unroll the recursion. Define the effective contraction factor

$$\rho_{\text{eff,wt}}(r,k) := \rho_{\text{eff}}(r)\,\big(1 + 2C\,\varepsilon_{\text{wt}}(k)\big),$$

so that

$$\begin{aligned}e_{k+1} &\le \rho_{\text{eff,wt}}(r,k)\,e_k \\ &\quad + C\,\delta_{\text{stat}}(n,\eta,K) + C\,\varepsilon_{\text{wt}}(k)^{\frac{1}{1-\alpha}} + C\,\varepsilon_{\text{wt}}(k)\,\varepsilon_{\mathcal{F}}.\end{aligned}$$

Unrolling this recursion yields

$$\begin{aligned}e_{k+1} &\le \left(\prod_{j=0}^{k} \rho_{\text{eff,wt}}(r,j)\right) e_0 \\ &\quad + C\sum_{i=0}^{k}\left(\prod_{j=i+1}^{k} \rho_{\text{eff,wt}}(r,j)\right)\Big[\delta_{\text{stat}}(n,\eta,K) + \varepsilon_{\text{wt}}(i)^{\frac{1}{1-\alpha}} \\ &\qquad + \varepsilon_{\text{wt}}(i)\,\varepsilon_{\mathcal{F}}\Big].\end{aligned}$$

By definition of the worst–case effective contraction factor $\rho_\star$, for all $0 \le j \le k \le K-1$,

$$\rho_{\mathrm{eff,wt}}(r,j) \;\le\; \rho_\star.$$

Using $\prod_{j=i+1}^{k} \rho_{\mathrm{eff,wt}}(r,j) \;\le\; \rho_\star^{k-i}$ and the geometric-series bound $\sum_{i=0}^{k} \rho_\star^{k-i} \le (1-\rho_\star)^{-1}$, we finally obtain, for a (possibly larger) constant $C > 0$,

$$\begin{aligned} e_{k+1} \le\; & \rho_\star^{k+1} e_0 \;+\; \frac{C}{1-\rho_\star}\,\delta_{\mathrm{stat}}(n,\eta,K) \\ & + \frac{C}{1-\rho_\star}\,\Delta_{\mathrm{wt}}(k+1,\alpha) \\ & + \frac{C\,\varepsilon_{\mathcal{F}}}{1-\rho_\star}\,\Delta_{\mathrm{wt}}(k+1,0). \end{aligned}$$

Reindexing $k := k+1$ yields the claimed bound.

□

G.3. Proof of Theorem 4

Proof of Theorem 4. We use the same induction structure as in the proof of Theorem 3. The base case $\widehat{Q}^{(0)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$ holds by Condition C12. Now fix $0 \le k \le K-1$ and assume $\widehat{Q}^{(k)} \in \mathbb{B}_{\mathcal{F}}(r, Q^\star_{\mathcal{F}})$. Throughout the proof, whenever we state that a bound holds with high probability, we mean that it holds uniformly for all $0 \le k \le K$.

Define the iterate error

$$e_k := \|\widehat{Q}^{(k)} - Q^\star_{\mathcal{F}}\|_{2,\mu^\star}.$$

The proof of Theorem 3 yields the recursion

$$\begin{aligned} e_{k+1} \le\; & \rho_{\mathrm{eff}}(r)\big(1 + 2C\,\varepsilon_{\mathrm{wt}}(k)\big)\, e_k \\ & + C\,\delta_{\mathrm{stat}}(n,\eta,K) + C\,\varepsilon_{\mathrm{wt}}(k)^{\frac{1}{1-\alpha}} \\ & + C\,\varepsilon_{\mathrm{wt}}(k)\big(1 + 2\rho_{\mathrm{eff}}(r)\big)\,\varepsilon_{\mathcal{F}}, \end{aligned}$$

for a constant $C < \infty$, ensuring the bound holds with probability at least $1 - \eta/2$.

Condition C11 implies that, for a (possibly different) constant $C < \infty$ and with probability at least $1 - \eta/2$,

$$\varepsilon_{\mathrm{wt}}(k) \;\le\; C\,e_k^{\kappa} \;+\; \delta_{\mathrm{wt}}(n,\eta,K).$$

Combining these inequalities, enlarging C if necessary to absorb all fixed multiplicative factors, and applying a union bound gives, with probability at least $1-\eta$,

$$\begin{aligned} e_{k+1} \le\; & \rho_{\mathrm{eff}}(r)\, e_k \;+\; 2\rho_{\mathrm{eff}}(r)\,\big(C e_k^{\kappa} + \delta_{\mathrm{wt}}(n,\eta,K)\big)\, e_k \\ & + C\Big(e_k^{\kappa} + \delta_{\mathrm{wt}}(n,\eta,K)\Big)^{\frac{1}{1-\alpha}} \\ & + C\big(e_k^{\kappa} + \delta_{\mathrm{wt}}(n,\eta,K)\big)\,\varepsilon_{\mathcal{F}} \;+\; C\,\delta_{\mathrm{stat}}(n,\eta,K). \end{aligned}$$

By the inequality $(a+b)^p \le 2^{p-1}(a^p + b^p)$ for $p \ge 1$, this yields, for a constant $C > 0$ (which may change from line to line),

$$\begin{aligned} e_{k+1} \le\; & \rho_{\mathrm{eff}}(r)\, e_k \;+\; C\,e_k^{1+\kappa} \;+\; C\,\delta_{\mathrm{wt}}(n,\eta,K)\, e_k \\ & + C\,e_k^{\frac{\kappa}{1-\alpha}} \;+\; C\Big(\delta_{\mathrm{stat}}(n,\eta,K) + \delta_{\mathrm{wt}}(n,\eta,K)^{\frac{1}{1-\alpha}}\Big) \\ & + C\,\varepsilon_{\mathcal{F}}\Big(e_k^{\kappa} + \delta_{\mathrm{wt}}(n,\eta,K)\Big). \end{aligned}$$

Absorbing constants and rearranging terms, there exists $B > 0$ (depending only on the structural constants) such that

$$\begin{aligned} e_{k+1} \le\; & \big(\rho_{\mathrm{eff}}(r) + B\,\delta_{\mathrm{wt}}(n,\eta,K)\big)\, e_k \\ & + B\Big(e_k^{1+\kappa} + e_k^{\frac{\kappa}{1-\alpha}} + \varepsilon_{\mathcal{F}}\, e_k^{\kappa}\Big) \\ & + B\Big(\delta_{\mathrm{stat}}(n,\eta,K) + \delta_{\mathrm{wt}}(n,\eta,K)^{\frac{1}{1-\alpha}} + \varepsilon_{\mathcal{F}}\,\delta_{\mathrm{wt}}(n,\eta,K)\Big). \end{aligned} \tag{25}$$

We now use induction to show ball invariance. Fix $r \in (0, r_{\max})$ and assume $e_k \le r$. Then

$$\begin{aligned} e_{k+1} \le\; & \big(\rho_{\mathrm{eff}}(r) + B\,\delta_{\mathrm{wt}}\big)\, r \\ & + B\Big(r^{1+\kappa} + r^{\frac{\kappa}{1-\alpha}} + \varepsilon_{\mathcal{F}}\, r^{\kappa}\Big) \\ & + B\Big(\varepsilon_{\mathrm{stat}}(n,\eta,K) + \delta_{\mathrm{wt}}^{\frac{1}{1-\alpha}} + \varepsilon_{\mathcal{F}}\,\delta_{\mathrm{wt}}\Big). \end{aligned}$$

Since r satisfies C12 and $e_k \le r$, we have

$$\begin{aligned} e_{k+1} \le\; & \big(\rho_{\mathrm{eff}}(r) + B\,\delta_{\mathrm{wt}}\big)\, r \\ & + B\Big(r^{1+\kappa} + r^{\frac{\kappa}{1-\alpha}} + \varepsilon_{\mathcal{F}}\, r^{\kappa}\Big) \\ & + B\Big(\varepsilon_{\mathrm{stat}}(n,\eta,K) + \delta_{\mathrm{wt}}^{\frac{1}{1-\alpha}} + \varepsilon_{\mathcal{F}}\,\delta_{\mathrm{wt}}\Big) \\ \le\; & \big(\rho_{\mathrm{eff}}(r) + B\,\delta_{\mathrm{wt}}\big)\, r + \big[1 - \rho_{\mathrm{eff}}(r) - B\,\delta_{\mathrm{wt}}\big]\, r \\ =\; & r. \end{aligned}$$

Consequently, by induction, all iterates remain in the ball of radius r, and the recursion in (25) holds with probability at least $1-\eta$, uniformly over $0 \le k \le K-1$.

We now unroll the recursion in (25). Recalling the definitions of $\Phi(e_j)$ and $\bar{\rho}$, we obtain

$$\begin{aligned} e_{k+1} \le\; & \bar{\rho}^{\,k+1} e_0 \;+\; B\sum_{j=0}^{k} \bar{\rho}^{\,k-j}\,\Phi(e_j) \;+\; B\Delta_{\mathrm{stat}} \sum_{j=0}^{k} \bar{\rho}^{\,k-j} \\ =\; & \bar{\rho}^{\,k+1} e_0 \;+\; B\sum_{j=0}^{k} \bar{\rho}^{\,k-j}\,\Phi(e_j) \;+\; B\Delta_{\mathrm{stat}}\,\frac{1-\bar{\rho}^{\,k+1}}{1-\bar{\rho}}. \end{aligned} \tag{26}$$

The bound then follows.

□